

Zero-Shot Linear Quadratic Regulation from Reward-Free Data: Structured Successor Features with Stability and Finite-Sample Guarantees

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Abstract

Reinforcement learning (RL) and adaptive dynamic programming (ADP) algorithms for the linear quadratic regulator (LQR) learn the optimal gain for *one* performance index: whenever the designer changes the weights (Q, R) , data must be re-collected or re-processed and the policy iteration re-run. Zero-shot RL, in contrast, learns from reward-free interaction a representation from which the optimal policy for *any* reward revealed at test time can be read out without further learning, but existing zero-shot methods rely on neural approximations and offer no stability or optimality guarantee. This paper brings zero-shot RL to linear optimal control. We show that for linear systems with quadratic costs the successor features of every linear policy are exact quadratic forms, that the policy-evaluation map $(Q, R) \mapsto Q$ -function kernel is linear and given by a closed-form successor operator, and that a fixed-size second-moment statistic of reward-free data is a sufficient zero-shot representation. On this basis we propose a zero-shot LQR algorithm for discrete-time systems: an offline phase compresses arbitrary reward-free trajectories into the representation, and a test-time phase maps any task (Q, R) , given explicitly or through reward labels, to a gain through a data-free policy iteration, or with no iteration at all through successor-feature generalized policy improvement (GPI). We prove that with noise-free persistently exciting data the returned gains are exactly the Riccati-optimal ones for all tasks simultaneously; that the GPI policy is exponentially stabilizing and no worse than any library policy; and that under sub-Gaussian process noise, with probability $1 - \delta$ and uniformly over a compact task set, the gains are stabilizing and their relative suboptimality is $O(\sigma_w \sqrt{\log(1/\delta)/(N\lambda_{PE})})$, where N is the number of samples and λ_{PE} the persistency-of-excitation level. A data-computable certificate verifies stability and suboptimality for each task. The framework extends to continuous-time systems through an integral-data Gram representation and a data-free Kleinman iteration. Simulations on the unstable batch reactor confirm machine-precision exactness over 200 random tasks, test-time cost independent of the data length (0.4 ms versus 14–359 ms for per-task off-policy Q-learning), and the predicted sample complexity.

Keywords: Zero-shot reinforcement learning, Linear quadratic regulator, Successor features, Adaptive dynamic programming, Data-driven control, Persistency of excitation

1. Introduction

Reinforcement learning (RL) and adaptive dynamic programming (ADP) have made it possible to solve linear quadratic regulation (LQR) problems without knowledge of the system matrices [Lewis and Vrabie \(2009\)](#); [Kiumarsi et al. \(2018\)](#); [Jiang and Jiang \(2017\)](#); [Lewis et al. \(2012\)](#). For discrete-time (DT) systems $x_{k+1} = Ax_k + Bu_k$, on- and off-policy Q-learning [Bradtke et al. \(1994\)](#); [Bradtke and Barto \(1996\)](#); [Lagoudakis and Parr \(2003\)](#); [Kiumarsi et al. \(2014\)](#); [Rizvi and Lin \(2019\)](#); [Zhang et al. \(2023\)](#) implement Hewer’s policy iteration [Hewer \(1971\)](#) from measured state–input data; for continuous-time (CT) systems $\dot{x} = Ax + Bu$, integral RL and off-policy ADP [Vrabie et al. \(2009\)](#); [Jiang and Jiang \(2012\)](#); [Vamvoudakis \(2017\)](#); [Possieri and Sassano \(2022\)](#) implement Kleinman’s algorithm [Kleinman \(1968\)](#). A parallel line of work on direct data-driven control [De Persis and Tesi \(2020\)](#); [van Waarde et al. \(2020\)](#); [Dörfler et al. \(2023\)](#); [Celi et al. \(2023\)](#), built on Willems’ fundamental lemma [Willems et al. \(2005\)](#),

and learning-theoretic analyses of LQR [Dean et al. \(2020\)](#); [Mania et al. \(2019\)](#); [Tu and Recht \(2019\)](#); [Krauth et al. \(2019\)](#); [Fazel et al. \(2018\)](#); [Abbasi-Yadkori and Szepesvári \(2011\)](#) characterize how much data such methods need.

All of these methods solve the problem for *one fixed* performance index. The weights (Q, R) appear in the Bellman equations that are solved from data, so when the designer changes them—to trade tracking accuracy against actuator effort, to adapt to a new mission phase, or simply during controller tuning—the policy iteration has to be run again on the stored data, or new data have to be collected when the method is on-policy. In practice the weights are rarely known in advance: they are tuned iteratively, and every tuning step then costs a full learning run.

The machine-learning community has studied the analogous question under the name *zero-shot RL* [Touati and Ollivier \(2021\)](#); [Touati et al. \(2023\)](#): an agent interacts with the environment *without any reward*, builds a representation, and at test time must output a near-optimal policy for *any* reward function, instantly and without further learning or planning in the environment. The two main families are successor features (SF)

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Dayan (1993); Barreto et al. (2017); Borsa et al. (2019), which assume rewards linear in known features and combine the SFs of several policies through generalized policy improvement (GPI), and forward–backward (FB) representations Touati and Ollivier (2021); Blier et al. (2021); Touati et al. (2023), which learn a low-rank factorization $M^{\pi_z}(s, a, s') \approx F(s, a, z)^\top B(s')$ of the successor measure together with a task-conditioned policy family π_z . Follow-up work improves representation learning Park et al. (2024) and robustness to low-quality data Jeon et al. (2024). These methods are evaluated on simulated locomotion and navigation benchmarks, rely on neural network approximation, and provide no closed-loop stability guarantee or error bound in terms of the amount and quality of data—the two properties that make a controller acceptable in control engineering.

Closer to control theory, multitask and meta-learning formulations of LQR have also been studied. Multitask LQG control Toso et al. (2026) learns a single shared stabilizing controller across a distribution of systems and objectives, with generalization bounds that grow with a bisimulation-based heterogeneity measure between tasks; the controller is common to all tasks and is therefore biased away from each task-specific optimum. Meta-learning approaches such as MAML-LQR Finn et al. (2017); Toso et al. (2024) instead learn an initialization from which a task-specific gain is reached after further policy-gradient adaptation on the new task, so a new task still requires online learning. A task-space-complexity framework for zero-shot generalization Cho et al. (2026) uses LQR as one of its running examples but measures transfer through a performance-divergence metric between source and target contexts rather than through a closed-form, exact policy map. None of these methods returns, from a single reward-free dataset and without further interaction or adaptation, the exact gain for every task in a continuum, or a stability certificate for it; this is what we do next.

To the authors’ knowledge, zero-shot RL in the representation-based sense above—a reward-free representation from which a near-optimal policy for *any* task is read out through a closed-form map, with provable stability and suboptimality guarantees—has not been formulated or analyzed for the LQR problem; the closest control-theoretic uses of the term “zero-shot” rely instead on a performance-divergence measure between a fixed source policy and unseen targets Cho et al. (2026), with no representation and no closed form. Conversely, none of the ADP or data-driven LQR methods above is zero-shot in the sense of amortizing learning across all performance indices. This paper connects the two lines. Our starting observation is that the linear–quadratic structure makes the objects that zero-shot RL approximates *exactly* finite-dimensional: with the quadratic features $\phi(z) = z \otimes z$, $z = [x^\top, u^\top]^\top$, every stage cost $x^\top Qx + u^\top Ru$ is linear in ϕ , the successor features of any linear policy are quadratic forms, and the map from the task to the policy’s Q-function kernel is a linear operator in closed form. The contributions are as follows.

1. **Problem and exact representation.** We formulate the

zero-shot LQR problem (Problem 4) and show (Proposition 6) that the successor features of LQR admit an exact forward–backward factorization, with a closed-form successor operator $\mathcal{L}_K$ that is linear in the task. A fixed-size second-moment statistic of reward-free data, whose size depends neither on the number of samples nor on the task, suffices to evaluate every policy on every task.

2. **Algorithms.** For DT systems we propose a two-phase algorithm: an offline phase that compresses arbitrary reward-free, off-policy trajectories (Algorithm 1), and a test-time phase that maps any task, given as weights or as reward labels, to a gain by a data-free policy iteration or, with no iteration, by SF-GPI over a policy library (Algorithm 2). For CT systems an integral-data Gram representation supports a data-free Kleinman iteration (Algorithm 3).
3. **Guarantees.** With noise-free persistently exciting (PE) data the returned gains equal the Riccati solutions for all tasks simultaneously, every iterate is stabilizing, and the iterates coincide with per-task off-policy Q-learning, which, unlike our method, needs quadratic rather than linear PE (Theorem 8). The SF-GPI policy is exponentially stabilizing and never worse than the best library policy (Theorem 9). Under sub-Gaussian process noise, with probability $1 - \delta$, the gains are stabilizing and the relative suboptimality is bounded uniformly over a compact task set by a quantity of order $\sigma_w \sqrt{\log(1/\delta)/(N\lambda_{\text{PE}})}$; a data-computable certificate verifies both properties for each individual task (Theorem 12). The CT counterpart is Theorem 15.
4. **Evidence.** On the open-loop unstable batch reactor we verify exactness over 200 random tasks, a test-time cost independent of the data length, the predicted $1/N$ decay of suboptimality for three PE levels, the near-optimality of zero-iteration SF-GPI, the CT extension, and task inference from noisy reward labels.

We state plainly what the structure implies. For LQR the zero-shot representation turns out to be equivalent to the least-squares estimate of the one-step transition operator $\Theta = [A \ B]$, so the test-time phase can be read as certainty-equivalent control computed on demand. We regard this as a finding rather than a weakness: it shows that the objects FB and SF methods learn with neural networks collapse, in the linear–quadratic case, to closed forms whose error can be controlled, and it gives the first finite-sample, closed-loop guarantees for zero-shot RL on continuous state and action spaces. The contributions that go beyond certainty equivalence are the operator viewpoint (Proposition 6), the zero-iteration SF-GPI controller with its stability proof, the uniform-over-tasks guarantee obtained from a *single* probabilistic event (no union bound over tasks), and the per-task data-computable certificate.

Organization. Section 2 states the problem. Section 3 derives the structured successor features. Section 4 presents the DT algorithm and its analysis, and Section 5 the CT extension. Section 6 reports simulations and Section 7 discusses limitations. Proofs are collected in the Appendix.

Notation. $A \succ 0$ ($\succeq 0$) denotes a positive (semi)definite matrix, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ the extreme eigenvalues of a symmetric matrix, $\|\cdot\|$ the Euclidean norm or the induced spectral norm, $\rho(\cdot)$ the spectral radius, and $\otimes$ the Kronecker product. $\text{vec}(\cdot)$ stacks columns, so that $\text{vec}(XYZ) = (Z^\top \otimes X) \text{vec}(Y)$ and $z^\top Wz = \text{vec}(W)^\top (z \otimes z)$. A matrix M is Schur if $\rho(M) < 1$ and Hurwitz if all its eigenvalues have negative real part. $\text{blkdiag}(\cdot)$ is a block-diagonal matrix. I_n is the $n \times n$ identity.

2. Problem formulation

2.1. System, tasks and costs

Consider the DT linear system

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k + w_k = \Theta z_k + w_k, \\ z_k &:= \begin{bmatrix} x_k \\ u_k \end{bmatrix} \in \mathbb{R}^d, \end{aligned} \quad (1)$$

with state $x_k \in \mathbb{R}^n$, input $u_k \in \mathbb{R}^m$, $d := n + m$, unknown $\Theta := [A \ B] \in \mathbb{R}^{n \times d}$, and process noise w_k (possibly zero). A task is a weight pair (Q, R) collected in

$$\begin{aligned} W &:= \text{blkdiag}(Q, R) \in \mathbb{R}^{d \times d}, \\ c(W) &:= \lambda_{\min}(W) = \min\{\lambda_{\min}(Q), \lambda_{\min}(R)\}. \end{aligned} \quad (2)$$

For a linear policy $u = -Kx$ let $\Pi_K := [I_n \ -K^\top]^\top \in \mathbb{R}^{d \times n}$, so that the closed-loop matrix is $A - BK = \Theta \Pi_K$. The set of stabilizing gains is $\mathcal{K}(\Theta) := \{K : \rho(\Theta \Pi_K) < 1\}$. For $K \in \mathcal{K}(\Theta)$ the value matrix $P_K(W)$ is the unique solution of the Lyapunov equation

$$\begin{aligned} P_K &= (A - BK)^\top P_K (A - BK) + Q + K^\top R K \\ &= (\Theta \Pi_K)^\top P_K (\Theta \Pi_K) + \Pi_K^\top W \Pi_K, \end{aligned} \quad (3)$$

and $x_0^\top P_K x_0 = \sum_{t \geq 0} (x_t^\top Q x_t + u_t^\top R u_t)$ along the noise-free closed loop. We measure performance by

$$J_W(K) := \text{tr } P_K(W), \quad (4)$$

the expected cost from $x_0 \sim \mathcal{N}(0, I_n)$, which also equals the average cost under unit-covariance process noise. The optimal value $P^*(W)$ is the stabilizing solution of the discrete algebraic Riccati equation (DARE)

$$\begin{aligned} P &= A^\top P A - A^\top P B (R + B^\top P B)^{-1} B^\top P A \\ &\quad + Q, \end{aligned} \quad (5)$$

with optimal gain $K^*(W) = (R + B^\top P^* B)^{-1} B^\top P^* A$ and $J_W^* = \text{tr } P^*(W)$. The associated Q-function kernel of a policy K is

$$\begin{aligned} H_K(W) &:= W + \Theta^\top P_K(W) \Theta = \begin{bmatrix} H^{xx} & H^{xu} \\ H^{ux} & H^{uu} \end{bmatrix}, \\ Q_K^W(x, u) &= z^\top H_K(W) z. \end{aligned} \quad (6)$$

2.2. Zero-shot LQR

Assumption 1 (Tasks). (A, B) is stabilizable. The task set $\mathcal{W}$ is a compact set of weights $W = \text{blkdiag}(Q, R)$ with $Q \succ 0, R \succ 0$, and $c_{\mathcal{W}} := \min_{W \in \mathcal{W}} c(W) > 0$.

Assumption 2 (Reward-free data). A dataset $\mathcal{D} = \{(x_k, u_k, x_{k+1})\}_{k=0}^{N-1}$ is generated by (1), possibly as several concatenated episodes, with inputs chosen by an arbitrary causal exploration rule (open-loop excitation, any behavior policy, or a mixture). No cost or reward is observed. Let $\mathcal{F}_k$ be the σ -algebra generated by $z_0, \dots, z_k, w_0, \dots, w_{k-1}$; then w_k is conditionally zero-mean given $\mathcal{F}_k$, and each coordinate of w_k is conditionally σ_w -sub-Gaussian.

Define the data matrices $Z := [z_0, \dots, z_{N-1}] \in \mathbb{R}^{d \times N}$ and $X^+ := [x_1, \dots, x_N] \in \mathbb{R}^{n \times N}$ and the empirical second moments

$$\hat{\Sigma}_{zz} := \frac{1}{N} Z Z^\top, \quad \hat{\Sigma}_{x^+z} := \frac{1}{N} X^+ Z^\top. \quad (7)$$

Definition 3 (PE level). The data are persistently exciting (PE) if $\text{rank } Z = d$, equivalently $\hat{\Sigma}_{zz} > 0$. The PE level is $\lambda_{\text{PE}} := \lambda_{\min}(\hat{\Sigma}_{zz})$.

Problem 4 (Zero-shot LQR). *Offline phase:* from $\mathcal{D}$ alone, compute a representation $\mathcal{R} = \mathcal{R}(\mathcal{D})$ whose size is independent of N and of any task. *Test phase:* for any $W \in \mathcal{W}$, revealed either explicitly or through reward labels $r_k = z_k^\top W z_k$ (possibly noisy) on $\mathcal{D}$, return a gain $\hat{K}(W) = \Gamma(\mathcal{R}, W)$ without interacting with the system and without reprocessing $\mathcal{D}$ (apart from one linear pass to infer W when only labels are given), such that

- (i) $\hat{K}(W) \in \mathcal{K}(\Theta)$ (closed-loop stability);
- (ii) $J_W(\hat{K}(W)) \leq (1 + \eta_N) J_W^*$ with $\sup_{W \in \mathcal{W}} \eta_N \rightarrow 0$ as $N \rightarrow \infty$ (uniform near-optimality);
- (iii) the computational cost of Γ is independent of N .

Classical off-policy Q-learning [Bradtke et al. \(1994\)](#); [Kiu-marsi et al. \(2014\)](#) satisfies (i)–(ii) for one task but violates (iii): every task requires a new policy iteration in which each policy-evaluation step solves a least-squares problem over all N samples. On-policy methods violate the offline requirement as well, since each task requires new interaction.

3. Structured successor features for LQR

Zero-shot RL rests on the successor representation [Dayan \(1993\)](#). For a policy π , the successor features with respect to a feature map ϕ are $\psi^\pi(z) = \sum_{t \geq 0} \phi(z_t)$ along the trajectory started at $z_0 = z$ [Barreto et al. \(2017\)](#). If the reward is linear in the features, $r = \phi^\top w$, then $Q_r^\pi = \psi^\pi{}^\top w$ for every w , so evaluating a known policy on a new task reduces to an inner product. FB representations [Touati and Ollivier \(2021\)](#) go one step further and learn a family of policies π_ζ together with a factorization $Q_r^{\pi_\zeta}(z) = F(z, \zeta)^\top \zeta$, where the task code $\zeta = \mathbb{E}[B(z)r(z)]$ is computed by a *backward* map; at test time π_ζ is read out directly.

For LQR we use the quadratic features and task code

$$\phi(z) := z \otimes z \in \mathbb{R}^{d^2}, \quad w := \text{vec}(W), \quad z^\top W z = \phi(z)^\top w. \quad (8)$$

Define the state–action transition matrix of the noise-free closed loop under K ,

$$G_K := \Pi_K \Theta \in \mathbb{R}^{d \times d}, \quad z_{t+1} = G_K z_t \text{ when } u_t = -Kx_t, \quad t \geq 1. \quad (9)$$

Lemma 5. *For every K , the nonzero eigenvalues of G_K coincide with those of $A - BK$, so $\rho(G_K) = \rho(A - BK)$. For $K \in \mathcal{K}(\Theta)$, the kernel (6) is the unique solution of*

$$H_K(W) = W + G_K^\top H_K(W) G_K, \quad (10)$$

and $P_K(W) = \Pi_K^\top H_K(W) \Pi_K$.

Proposition 6 (Exact successor features and successor operator). *Let $K \in \mathcal{K}(\Theta)$ and consider the noise-free closed loop. Then:*

- (a) *the successor features of K are $\psi^K(z) = (I_d - G_K \otimes G_K)^{-1}(z \otimes z)$, a quadratic form in z ;*
- (b) *for every task, $\mathbf{Q}_K^W(z) = \psi^K(z)^\top \text{vec}(W)$ and $\text{vec}(H_K(W)) = \mathcal{L}_K \text{vec}(W)$ with the successor operator*

$$\mathcal{L}_K := (I_d - G_K^\top \otimes G_K^\top)^{-1}; \quad (11)$$

- (c) *G_K , and hence $\mathcal{L}_K$ and ψ^K , depend on the data-generating system only through Θ .*

Proposition 6 says that in LQR the Q-function of any linear policy is *linear in the task*, with an explicit operator. It maps onto the FB template: the backward map is the least-squares projection of rewards on ϕ ,

$$\hat{w} = \left(\sum_k \phi(z_k) \phi(z_k)^\top \right)^\dagger \sum_k \phi(z_k) r_k, \quad (12)$$

which recovers w exactly from noise-free labels whenever the quadratic features of the data span the symmetric matrices; the forward map is $F(z, W) = \psi^{K^*(W)}(z)$, so that $\mathbf{Q}_W^*(z) = F(z, W)^\top \text{vec}(W)$. Unlike generic FB, neither map has to be approximated: ψ^K is available in closed form once Θ is, and $K^*(W)$ is reached by policy iteration in which every evaluation is (10). Proposition 6(c) identifies what the offline phase must learn: the one-step transition operator Θ . When the data are PE,

$$\hat{\Theta} := \hat{\Sigma}_{x^+z} \hat{\Sigma}_{zz}^{-1} = X^+ Z^\top (Z Z^\top)^{-1} \quad (13)$$

is its least-squares estimate. The pair $\mathcal{R} := (\hat{\Sigma}_{zz}, \hat{\Sigma}_{x^+z})$ has $d^2 + nd$ entries regardless of N and is therefore an admissible zero-shot representation. We write $\hat{G}_K := \Pi_K \hat{\Theta}$, $\hat{\mathcal{L}}_K$, $\hat{H}_K(W)$ and $\hat{P}_K(W)$ for the quantities computed with $\hat{\Theta}$ in place of Θ .

Remark 7 (Why linear rather than quadratic regression). Off-policy Q-learning [Bradtke et al. \(1994\)](#); [Kiumarsi et al. \(2014\)](#) evaluates K by regressing $z_k^\top W z_k$ on $\phi(z_k) - \phi(\Pi_K x_{k+1})$, which needs the *quadratic* PE condition $\text{rank}[\phi(z_0), \dots, \phi(z_{N-1})] = d(d+1)/2$, and therefore at least $d(d+1)/2$ samples. Estimating the linear operator Θ needs only $\text{rank } Z = d$. Under process noise the quadratic regression is moreover biased by the term $\text{tr}(H \text{Cov}(w))$, whereas (13) is consistent.

Algorithm 1 Offline phase: reward-free representation (DT)

Input: reward-free data $\{(x_k, u_k, x_{k+1})\}_{k=0}^{N-1}$; optional anchor tasks $W_1, \dots, W_M$

- 1: $\hat{\Sigma}_{zz} \leftarrow \frac{1}{N} \sum_k z_k z_k^\top$, $\hat{\Sigma}_{x^+z} \leftarrow \frac{1}{N} \sum_k x_{k+1} z_k^\top$ ▷ streaming, $O(d^2)$ memory
- 2: check PE: $\lambda_{\text{PE}} = \lambda_{\min}(\hat{\Sigma}_{zz}) > 0$; $\hat{\Theta} \leftarrow \hat{\Sigma}_{x^+z} \hat{\Sigma}_{zz}^{-1}$
- 3: **for** $i = 1, \dots, M$ **do** ▷ optional policy library
- 4: $K_i \leftarrow \text{Mode PI of Algorithm 2 (or DARE of } \hat{\Theta}) \text{ for } W_i$
- 5: $\hat{\mathcal{L}}_{K_i} \leftarrow (I - \hat{G}_{K_i}^\top \otimes \hat{G}_{K_i}^\top)^{-1}$
- 6: **return** $\mathcal{R} = (\hat{\Sigma}_{zz}, \hat{\Sigma}_{x^+z})$ and library $\{(K_i, \hat{\mathcal{L}}_{K_i})\}_{i=1}^M$; the raw data may be discarded

4. Zero-shot LQR for discrete-time systems

4.1. Algorithms

The offline phase (Algorithm 1) accumulates the second moments. It optionally precomputes a *policy library*: M gains $K_1, \dots, K_M$ that stabilize $\hat{\Theta}$ (for instance $\hat{K}(W_i)$ for anchor tasks $W_1, \dots, W_M$) together with their successor operators $\hat{\mathcal{L}}_{K_i}$. The test phase (Algorithm 2) supports two modes.

Mode PI (policy iteration in representation space). Starting from a gain $K_0 \in \mathcal{K}(\hat{\Theta})$, iterate

$$\hat{H}_j = W + \hat{G}_{K_j}^\top \hat{H}_j \hat{G}_{K_j}, \quad K_{j+1} = (\hat{H}_j^{uu})^{-1} \hat{H}_j^{ux}. \quad (14)$$

Each step solves a $d \times d$ Lyapunov equation and touches no data. The initial gain can be taken from the library as $K_0 = \arg \min_{K_i} \text{tr}(\Pi_{K_i}^\top \hat{H}_{K_i}(W) \Pi_{K_i})$, which costs M matrix–vector products with the precomputed $\hat{\mathcal{L}}_{K_i}$. This removes the usual requirement of a task-specific initial stabilizing gain.

Mode GPI (no iteration). For each library policy compute $\hat{H}_i = \hat{H}_{K_i}(W)$ from $\text{vec}(\hat{H}_i) = \hat{\mathcal{L}}_{K_i} \text{vec}(W)$, the improved gain $K_i^+ = (\hat{H}_i^{uu})^{-1} \hat{H}_i^{ux}$, and the value kernel $S_i = \hat{H}_i^{xx} - \hat{H}_i^{xu} K_i^+$. Apply the switched linear policy

$$\pi_W^{\text{GPI}}(x) = -K_{i^*(x)}^+ x, \quad i^*(x) \in \arg \min_i x^\top S_i x. \quad (15)$$

This is exactly GPI [Barreto et al. \(2017\)](#), $\pi(x) \in \arg \min_u \min_i \mathbf{Q}_{K_i}^W(x, u)$, written in closed form. Its test-time cost is M products of a $d^2 \times d^2$ matrix with a vector.

4.2. Exact zero-shot optimality with noise-free data

Theorem 8 (Exactness). *Let $w_k \equiv 0$, let the data be PE, and let (A, B) be stabilizable. Then $\hat{\Theta} = \Theta$, and for every W with $Q > 0$, $R > 0$ and every $K_0 \in \mathcal{K}(\Theta)$ the iteration (14) satisfies:*

- (a) $K_j \in \mathcal{K}(\Theta)$ for all j ;
- (b) $P^*(W) \leq P_{K_{j+1}}(W) \leq P_{K_j}(W)$;
- (c) $K_j \rightarrow K^*(W)$ and $P_{K_j}(W) \rightarrow P^*(W)$ quadratically.

If in addition the quadratic features of the data span the symmetric matrices, the iterates coincide with those of per-task off-policy Q-learning [Bradtke et al. \(1994\)](#); [Kiumarsi et al. \(2014\)](#) run on $\mathcal{D}$ with the same W and K_0 , and the backward map (12) recovers W exactly from noise-free labels.

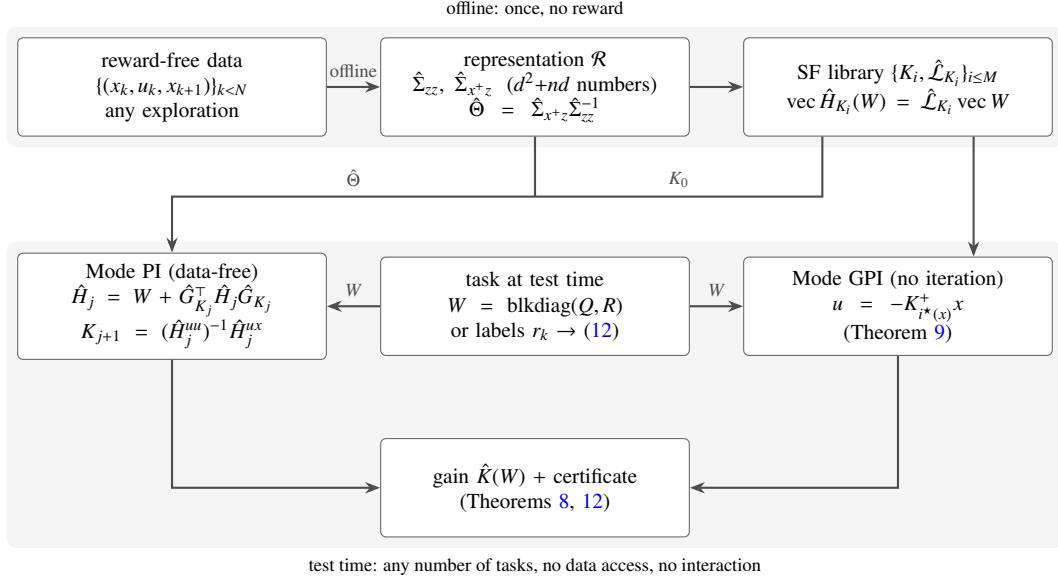


Figure 1: Zero-shot LQR. The offline phase compresses reward-free data into a fixed-size representation and, optionally, an SF library. At test time any task is mapped to a gain without touching the data or the plant.

Algorithm 2 Test phase: zero-shot gain for a new task (DT)

Input: representation $\mathcal{R}$, library, task W or labels $\{r_k\}$; accuracy ϵ_{tol} ; optional noise level $\bar{\epsilon}$

- 1: **if** only labels are given **then**
- 2: $W \leftarrow$ backward map (12); project Q, R onto $\{\geq \epsilon I\}$
- 3: **for** $i = 1, \dots, M$ **do**
- 4: $\text{vec}(\hat{H}_i) \leftarrow \hat{\mathcal{L}}_{K_i} \text{vec}(W)$
- 5: **if** Mode GPI **then**
- 6: **return** switched policy (15)
- 7: $K_0 \leftarrow \arg \min_i \text{tr}(\Pi_{K_i}^\top \hat{H}_i \Pi_{K_i})$, $j \leftarrow 0$
- 8: **repeat**
- 9: $\hat{H}_j \leftarrow$ solve $\hat{H}_j = W + \hat{G}_{K_j}^\top \hat{H}_j \hat{G}_{K_j}$; $K_{j+1} \leftarrow (\hat{H}_j^{uu})^{-1} \hat{H}_j^{ux}$;
 $j \leftarrow j + 1$
- 10: **until** $\|K_j - K_{j-1}\| < \epsilon_{\text{tol}}$
- 11: $\hat{K} \leftarrow K_j$, $\hat{P} \leftarrow \Pi_{\hat{K}}^\top \hat{H}_{\hat{K}} \Pi_{\hat{K}}$
- 12: **certificate:** if $\bar{\epsilon} \leq c(W)/(12\|\hat{P}\|)$, report “stabilizing,
 $J_W(\hat{K}) \leq (1 + 20\bar{\epsilon}\|\hat{P}\|/c(W))J_W^*$ ” ▷ Theorem 12
- 13: **return** $\hat{K}$

The theorem holds for all tasks at once: a single reward-free dataset, compressed once, solves every LQR problem on the plant exactly. Only linear PE is needed, which requires $N \geq d$ samples, compared with $N \geq d(d+1)/2$ for per-task Q-learning.

Theorem 9 (Zero-iteration SF-GPI). *Let $\hat{\Theta} = \Theta$ (e.g., under the hypotheses of Theorem 8), let $K_1, \dots, K_M \in \mathcal{K}(\Theta)$, and let W satisfy $Q > 0, R > 0$. Define $V(x) := \min_i x^\top S_i x$. Then along the closed loop $x_{t+1} = Ax_t + B\pi_W^{\text{GPI}}(x_t)$:*

- (a) $V(x_{t+1}) \leq V(x_t) - x_t^\top Q x_t$, and $\lambda_{\min}(Q)\|x\|^2 \leq V(x) \leq \max_i \|S_i\| \|x\|^2$; hence the origin is globally exponentially stable, with $V(x_t) \leq (1 - \lambda_{\min}(Q)/\max_i \|S_i\|)^t V(x_0)$;
- (b) the accumulated cost satisfies $\sum_{t \geq 0} (x_t^\top Q x_t + u_t^\top R u_t) \leq$

$$V(x_0) \leq \min_i x_0^\top P_{K_i}(W) x_0 \text{ for every } x_0.$$

For $M = 1$, SF-GPI reduces to one Hwuer improvement step from K_1 . For $M > 1$ it is no worse, from every initial state, than the best library policy for that state, without solving any equation at test time.

4.3. Finite-sample guarantees with noisy data

Let $\bar{z}^2 \geq \lambda_{\max}(\hat{\Sigma}_{zz})$ and $\lambda_0 \leq \lambda_{\text{PE}}$ be deterministic levels, and define

$$\bar{\epsilon}_N(\delta) := \sqrt{\frac{4n\sigma_w^2}{N\lambda_0} \left(\frac{d}{2} \log(1 + \bar{z}^2/\lambda_0) + \log \frac{n}{\delta} \right)}. \quad (16)$$

Lemma 10 (Representation error). *Under Assumption 2, for any $\delta \in (0, 1)$ the event*

$$\mathcal{E} := \{\lambda_0 I \leq \hat{\Sigma}_{zz} \leq \bar{z}^2 I\} \cap \{\|\hat{\Theta} - \Theta\| > \bar{\epsilon}_N(\delta)\}$$

has probability at most δ .

Lemma 11 (One-sided perturbation). *Let $\Theta_1, \Theta_2 \in \mathbb{R}^{n \times d}$ with $\|\Theta_1 - \Theta_2\| \leq \epsilon$, let W satisfy $c = c(W) > 0$, let K stabilize Θ_1 , and let P_1 be its value matrix on Θ_1 with $p_1 = \|P_1\|$. If $\epsilon \leq c/(6p_1)$, then K stabilizes Θ_2 , and its value matrix on Θ_2 satisfies $P_2 \leq (1 + 6\epsilon p_1/c)P_1$.*

Theorem 12 (Uniform zero-shot guarantee and certificate). *Let Assumptions 1–2 hold, fix $\delta \in (0, 1)$, and let $\hat{K}(W), \hat{P}(W)$ be the output of Algorithm 2 (Mode PI run to convergence) with $\bar{\epsilon} = \bar{\epsilon}_N(\delta)$. With probability at least $1 - \delta$, whenever $\lambda_0 I \leq \hat{\Sigma}_{zz} \leq \bar{z}^2 I$, the following hold simultaneously for all $W \in \mathcal{W}$.*

- (a) **Certificate.** *If $\bar{\epsilon}_N(\delta) \leq c(W)/(12\|\hat{P}(W)\|)$, then $\hat{K}(W) \in \mathcal{K}(\Theta)$ and*

$$P^*(W) \leq P_{\hat{K}}(W) \leq (1 + 20\bar{\epsilon}_N \frac{\|\hat{P}\|}{c(W)})P^*(W), \quad (17)$$

$$\|\hat{P} - P^*\| \leq 14\bar{\epsilon}_N \frac{\|\hat{P}\|^2}{c(W)}, \quad \|\hat{K} - K^*\| \leq \frac{\|\hat{P}\|}{c(W)} \sqrt{30\bar{\epsilon}_N}.$$

(b) Uniform rate. Let $\bar{p}_W := \max_{W \in \mathcal{W}} \|P^*(W)\|$. If $\bar{\varepsilon}_N(\delta) \leq c_W/(18\bar{p}_W)$, then every $W \in \mathcal{W}$ satisfies the certificate in (a), and

$$\sup_{W \in \mathcal{W}} \frac{J_W(\hat{K}(W)) - J_W^*}{J_W^*} \leq 27 \frac{\bar{p}_W}{c_W} \bar{\varepsilon}_N(\delta) \quad (18)$$

which is

$$O\left(\frac{\bar{p}_W}{c_W} \sigma_w \sqrt{\frac{n(d \log(1 + \bar{z}^2/\lambda_0) + \log(n/\delta))}{N \lambda_0}}\right). \quad (19)$$

Remark 13 (Role of data and excitation). The bound (18) decreases as $1/\sqrt{N\lambda_0}$: halving the error requires either four times more data or four times more excitation. Since λ_{PE} and $\bar{z}^2$ are computed from $\hat{\Sigma}_{zz}$, both the bound and the certificate in (a) can be evaluated from the representation alone, given σ_w . Only one probabilistic event is involved, so the guarantee holds for all tasks at once, with no union bound over $\mathcal{W}$, which is what makes it a zero-shot guarantee.

Remark 14 (Tightness). The rate in (18) is linear in $\bar{\varepsilon}_N$, which is enough for a clean, fully explicit and data-computable certificate. Local analyses of certainty-equivalent LQR Mania et al. (2019) show that the suboptimality is in fact quadratic in the model error for small errors, i.e. $O(1/(N\lambda_{PE}))$. Our experiments (Fig. 3) display this faster rate. Obtaining a uniform-over-tasks version of the quadratic rate with explicit constants is left for future work.

5. Extension to continuous-time systems

Consider $\dot{x} = Ax + Bu$ with unknown (A, B) , cost $\int_0^\infty (x^\top Qx + u^\top Ru) dt$, and optimal gain $K^*(W) = R^{-1}B^\top P^*$, where P^* solves the continuous-time ARE. Kleinman's algorithm Kleinman (1968) iterates $(A - BK_j)^\top P_j + P_j(A - BK_j) + Q + K_j^\top RK_j = 0$, $K_{j+1} = R^{-1}B^\top P_j$. Following Jiang and Jiang (2012), apply an exploratory input $u = -K_b x + e(t)$ on $[t_0, t_l]$ and, for a partition $t_0 < t_1 < \dots < t_l$, form the rows ($\bar{x}$ denotes the vector with $x^\top Px = \bar{x}^\top \text{svec}(P)$)

$$\begin{aligned} \delta_{xx} &= [\bar{x}^\top]_{t_{l-1}}^{t_l}, \\ I_{xx} &= \left[\int_{t_{l-1}}^{t_l} (x \otimes x)^\top d\tau \right]_i, \\ I_{xu} &= \left[\int_{t_{l-1}}^{t_l} (x \otimes u)^\top d\tau \right]_i. \end{aligned}$$

For a gain K_j , the Kleinman step is equivalent to the linear equation Jiang and Jiang (2012)

$$\begin{aligned} & \underbrace{[\delta_{xx}, -2I_{xx}(I_n \otimes K_j^\top R) - 2I_{xu}(I_n \otimes R)]}_{=DT(K_j, R)} \begin{bmatrix} \text{svec}(P_j) \\ \text{vec}(K_{j+1}) \end{bmatrix} \\ &= \underbrace{-I_{xx} \text{vec}(Q + K_j^\top RK_j)}_{=Ds(K_j, W)}, \end{aligned} \quad (20)$$

where $D := [\delta_{xx}, I_{xx}, I_{xu}]$ is the raw data matrix and $T(\cdot)$, $s(\cdot)$ are known task-dependent linear maps that do not depend on

Algorithm 3 Zero-shot LQR for CT systems

Input: offline: integral data $(\delta_{xx}, I_{xx}, I_{xu})$; a gain K_0 that stabilizes (A, B)

- 1: **offline:** $\mathcal{R}_{CT} \leftarrow D^\top D$; discard the data
- 2: **test** (any W): $j \leftarrow 0$
- 3: **repeat**
- 4: $T \leftarrow T(K_j, R)$, $s \leftarrow s(K_j, W)$; solve $T^\top \mathcal{R}_{CT} T \theta = T^\top \mathcal{R}_{CT} s$; $K_{j+1} \leftarrow$ the K -block of θ ; $j \leftarrow j + 1$
- 5: **until** $\|K_j - K_{j-1}\| < \epsilon_{\text{tol}}$
- 6: **return** $\hat{K}(W) = K_j$

the data. Consequently the normal equations of (20) depend on the data only through the Gram matrix

$$\mathcal{R}_{CT} := D^\top D \in \mathbb{R}^{p \times p}, \quad p = \frac{n(n+1)}{2} + n^2 + nm, \quad (21)$$

whose size does not depend on the number of intervals l . Algorithm 3 is the resulting zero-shot method.

Theorem 15 (CT exactness). Suppose the data are noise-free and satisfy the rank condition $\text{rank}[I_{xx}, I_{xu}] = n(n+1)/2 + nm$ of Jiang and Jiang (2012). Then for every W with $Q > 0$, $R > 0$ and a single gain K_0 that makes $A - BK_0$ Hurwitz, the iterates of Algorithm 3 coincide with Kleinman's iterates for W ; hence every K_j is stabilizing and $K_j \rightarrow K^*(W)$ quadratically. The test-time cost per iteration is $O(p^3)$, independent of l .

Unlike the DT case, the CT method needs a known stabilizing K_0 because Kleinman's iteration must be initialized with one. The same K_0 , however, serves all tasks, since stability of $A - BK_0$ does not depend on W .

6. Simulation results

6.1. Setup

We use Rosenbrock's batch reactor Walsh et al. (2002); De Persis and Tesi (2020),

$$\begin{aligned} A_c &= \begin{bmatrix} 1.38 & -0.2077 & 6.715 & -5.676 \\ -0.5814 & -4.29 & 0 & 0.675 \\ 1.067 & 4.273 & -6.654 & 5.893 \\ 0.048 & 4.273 & 1.343 & -2.104 \end{bmatrix}, \\ B_c &= \begin{bmatrix} 0 & 0 \\ 5.679 & 0 \\ 1.136 & -3.146 \\ 1.136 & 0 \end{bmatrix}, \end{aligned}$$

which is open-loop unstable (eigenvalues 1.99, 0.064, -5.06, -8.67), with $n = 4$, $m = 2$, $d = 6$. The DT model is the zero-order-hold discretization with sampling time $h = 0.1$ s ($\rho(A) = 1.22$). DT data consist of episodes of 10 steps from $x_0 \sim \mathcal{N}(0, I)$ with open-loop inputs $u_k \sim \mathcal{N}(0, \sigma_e^2 I)$; no feedback, no stabilizing gain, and no reward are used. Test tasks are drawn as $Q = U \text{diag}(10^{a_i}) U^\top$, $R = V \text{diag}(10^{b_i}) V^\top$ with random orthogonal U, V and $a_i, b_i \sim \mathcal{U}(-1, 1)$, so that the weights span two orders of magnitude and are not diagonal.

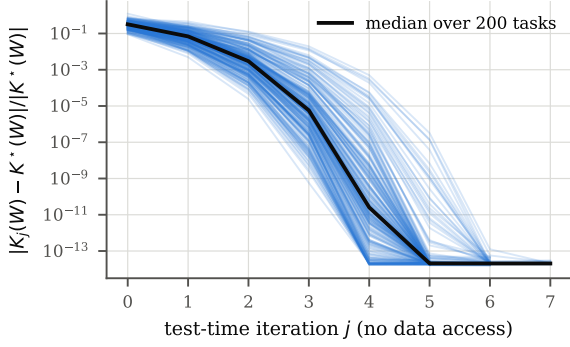


Figure 2: E1: relative gain error of the data-free test-time policy iteration for 200 random tasks from one reward-free dataset ($N = 200$). Thin lines: individual tasks; thick line: median.

Table 1: E1: per-task computation time (mean over 20 tasks) and the largest gap between the gains of the two methods.

N	offline (ms)	zero-shot (ms)	off-policy QL (ms)	$\max \ \hat{K} - K_{QL}\ $
200	0.05	0.37	14.5	3.1×10^{-13}
1000	0.08	0.39	73.3	1.4×10^{-12}
5000	0.14	0.39	358.5	2.2×10^{-13}

Ground truth is computed with the DARE/CARE of the true model. All code and data needed to reproduce every number are in the accompanying repository (code/experiments.py). [\[TODO: add repository link\]](#)

6.2. E1: exactness and test-time cost (DT, noise-free)

With $N = 200$ samples (20 episodes, $\sigma_e = 1$, $\lambda_{PE} = 0.138$), Mode PI was run for 200 random tasks from the same initial gain. All iterates of all tasks were stabilizing, the largest relative gain error was 5.6×10^{-14} (machine precision), and convergence took 5.6 iterations on average (at most 7). Fig. 2 shows the quadratic convergence. Table 1 compares the per-task test-time cost with per-task off-policy Q-learning on the same data: the two methods return the same gains up to 1.4×10^{-12} , but the cost of Q-learning grows linearly with N while ours does not depend on N .

6.3. E2: sample complexity and PE level (DT, noisy)

Process noise with $\sigma_w = 0.05$ was added and the exploration level was varied over $\sigma_e \in \{0.3, 1, 3\}$, which yields PE levels $\lambda_{PE} \approx 0.09, 0.15, 0.33$. For each N from 100 to 3×10^4 we averaged five independent datasets and evaluated 100 tasks. Fig. 3(a) shows that $\|\hat{\Theta} - \Theta\|$ decays as $N^{-1/2}$ and decreases with the PE level, as predicted by Lemma 10; the high-probability bound $\bar{\epsilon}_N(\delta)$ ($\delta = 0.05$) is valid and conservative by a factor of about 6–10. Fig. 3(b) reports the worst-task and median-task relative suboptimality. All $3 \times 6 \times 5 \times 100 = 9000$ returned gains stabilized the true plant. The worst-task suboptimality decays from 6.5×10^{-2} , 1.8×10^{-2} and 5.3×10^{-3} at $N = 100$ to 2.4×10^{-4} , 4.8×10^{-5} and 4.1×10^{-6} at $N = 3 \times 10^4$ for $\sigma_e = 0.3, 1, 3$ respectively, with a slope close to -1 , i.e. the quadratic regime discussed after Theorem 12.

Table 2: E3: relative suboptimality (median / worst over 40 tasks).

M	best library policy	SF-GPI	GPI-init + 1 step
1	0.220 / 1.460	0.0071 / 0.377	0.0071 / 0.377
2	0.131 / 1.218	0.0047 / 0.258	0.0055 / 0.261
4	0.079 / 0.519	0.0019 / 0.056	0.0021 / 0.077
8	0.063 / 0.311	0.0014 / 0.037	0.0014 / 0.080
16	0.041 / 0.142	0.0007 / 0.013	0.0011 / 0.030

The certificate of Theorem 12(a) is sound but conservative: it never failed when it was issued, but with the true error $\|\hat{\Theta} - \Theta\|$ in place of $\bar{\epsilon}_N$ it covered only 3%, 9% and 28% of the tasks at $N = 3 \times 10^4$, and none when $\bar{\epsilon}_N$ was used. The reason is the factor $\|\hat{P}\|/c(W)$, which ranges from 23 for $Q = I, R = I$ to more than 10^3 for the most ill-conditioned random tasks. For $Q = I, R = I$ and $\sigma_e = 3$ the certificate with $\bar{\epsilon}_N$ would be obtained for $N \gtrsim 1.7 \times 10^5$. Tightening the constants is discussed in Section 7.

6.4. E3: zero-iteration SF-GPI

With a noise-free representation ($N = 1000$), libraries of $M \in \{1, 2, 4, 8, 16\}$ gains optimal for random anchor tasks were built, and 40 new test tasks were solved in Mode GPI; the closed-loop cost of the switched policy (15) was obtained by simulation from 10 initial states. Fig. 4 and Table 2 compare SF-GPI with the best single library policy and with one Newton step from the library initialization. GPI improves on the best library policy by a factor of 30–60 in median suboptimality and reaches 0.07% with $M = 16$, at a test-time cost of 0.03–0.12 ms and without solving any equation. Every GPI closed loop was stable, as guaranteed by Theorem 9.

6.5. E4: continuous time

The CT reactor was excited with $u = -K_b x + e(t)$, where K_b is one stabilizing gain and e a sum of ten sinusoids per channel, over 2 s split into $l = 100$ intervals; the rank condition holds ($18 = 18$). The Gram representation (21) is 34×34 . For 50 random tasks started from the same $K_0 = K_b$, Algorithm 3 converged in 6.8 iterations on average, all iterates were Hurwitz-stabilizing, and the gains agree with per-task off-policy ADP Jiang and Jiang (2012) to 8×10^{-11} . The remaining 2×10^{-4} relative error with respect to the CARE solution comes from the numerical quadrature of the integrals, not from the method (Fig. 5(a)). Fig. 5(b) shows closed-loop responses for three very different tasks computed from the same data.

6.6. E5: tasks specified by reward labels

Instead of (Q, R) , the learner received labels $r_k = z_k^\top W z_k + v_k$ on the $N = 1000$ stored samples, with v_k Gaussian of standard deviation equal to a fraction of $\text{std}(r)$. Table 3 shows that noise-free labels identify W exactly, while 1% label noise causes a 2% weight error and a median suboptimality of 5×10^{-5} . The LQR solution is insensitive to moderate errors in the weights, so reward-label task specification is practical.

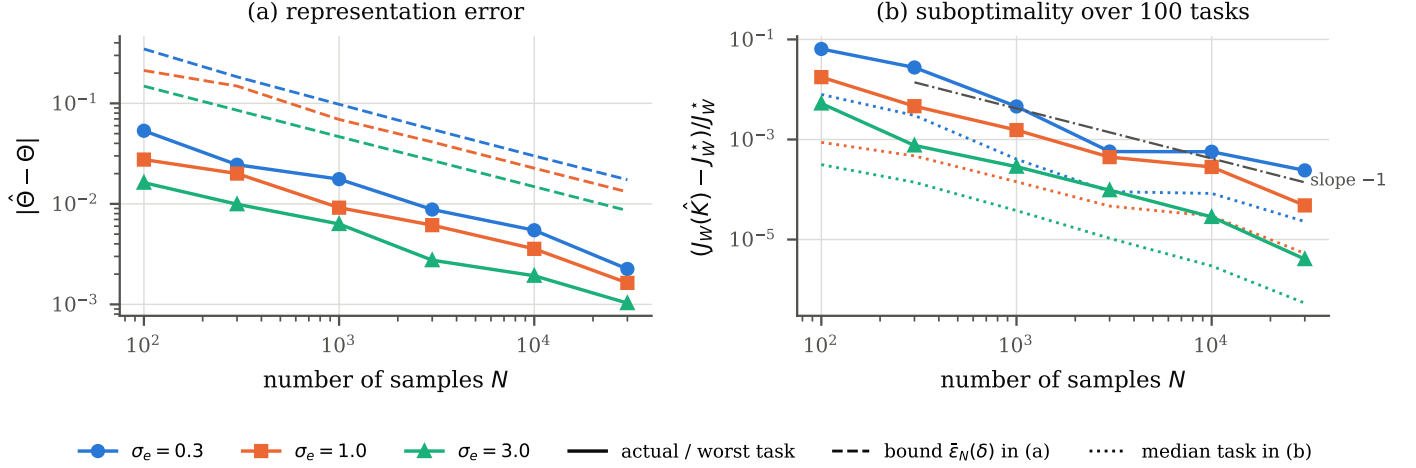


Figure 3: E2: (a) representation error versus N for three excitation levels (solid: observed, dashed: bound $\bar{\epsilon}_N(\delta)$ of Lemma 10); (b) worst-task (solid) and median-task (dotted) relative suboptimality over 100 tasks.

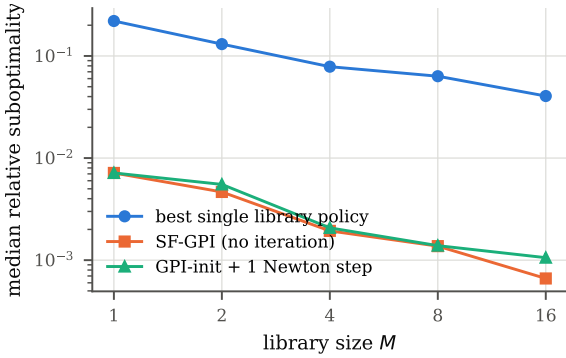


Figure 4: E3: median relative suboptimality over 40 unseen tasks versus library size.

Table 3: E5: task inference from noisy reward labels (50 tasks).

label noise	median $\frac{\ \hat{W} - W\ _F}{\ W\ _F}$	median subopt.	worst subopt.
0	1.7×10^{-14}	≈ 0	3.3×10^{-14}
1%	0.020	4.6×10^{-5}	3.0×10^{-3}
10%	0.204	5.5×10^{-3}	0.168
100%	2.12	0.191	1.76

7. Discussion and limitations

Relation to certainty equivalence. In the linear-quadratic case the minimal zero-shot representation is equivalent to the least-squares estimate of Θ , and Mode PI returns the certainty-equivalent gain. This is not a limitation of the construction but a property of the problem. What the zero-shot viewpoint adds is (i) the task-linear successor operator, which makes policy evaluation for a new task a matrix-vector product and enables zero-iteration GPI; (ii) guarantees that hold for all tasks simultaneously with a per-task data-computable certificate; and (iii) a template for the nonlinear case, where no closed-form model is available.

Conservatism. The constants in Theorem 12 are explicit but

loose, mainly through the factor $\|\hat{P}\|/c(W)$ and the linear dependence on $\bar{\epsilon}_N$. Sharper certificates could use the quadratic perturbation theory of Mania et al. (2019), self-normalized bounds with ℓ_2 instead of per-row union bounds Simchowicz et al. (2018), or robust synthesis over the confidence set in the spirit of Dean et al. (2020).

Assumptions. We assume full-state measurement, $Q > 0$, and a known noise level σ_w for the certificate. The extension to detectable $(A, Q^{1/2})$, to output feedback Rizvi and Lin (2019), and to H_∞ or tracking tasks, where the task additionally contains a reference or an attenuation level, are natural next steps.

Nonlinear systems. For $\dot{x} = f(x, u)$ the same two-phase structure applies with a lifted feature map ϕ (e.g., Koopman observables Korda and Mezić (2018) or neural features) and costs linear in ϕ . The successor operator then becomes approximate, and the analysis requires uniformly ultimately bounded (UUB) arguments of the ADP type. This is the most important open direction.

8. Conclusion

We formulated zero-shot reinforcement learning for linear quadratic regulation and showed that successor features and the forward-backward factorization become exact and finite-dimensional in this setting. The resulting algorithm learns once from reward-free, off-policy data and returns, for any quadratic performance index revealed at test time, a stabilizing gain whose suboptimality is bounded uniformly over tasks in terms of the amount of data and the persistency-of-excitation level, or a zero-iteration GPI controller with a Lyapunov stability guarantee. Simulations on the unstable batch reactor confirm exactness with noise-free data, test-time cost independent of data length, the predicted sample complexity, and the effectiveness of SF-GPI.

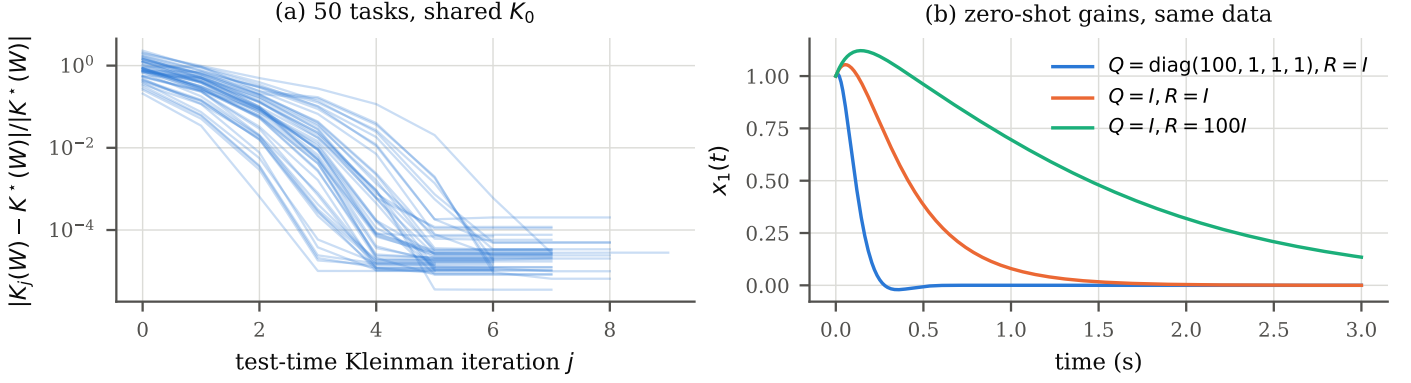


Figure 5: E4 (CT): (a) data-free Kleinman iteration for 50 tasks from one stabilizing gain; (b) first state of the closed loop under zero-shot gains for three tasks.

CRedit authorship contribution statement

Quang Huy Dao: Conceptualization, Methodology, Software, Writing – original draft. **Khanh Thai:** Conceptualization, Methodology, Software, Writing – original draft. **Phuong Nam Dao:** Supervision, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proofs

This appendix proves every result of the paper step by step. Section [Appendix A.1](#) collects eight elementary facts that are used repeatedly; each later step states which fact it uses.

Appendix A.1. Elementary facts

Throughout, $X \leq Y$ means that $Y - X$ is positive semidefinite (the Loewner order).

(F1) Loewner order. For symmetric X, Y and any matrix T of compatible size:

- (i) $X \leq Y \Rightarrow T^\top X T \leq T^\top Y T$, because $v^\top T^\top (Y - X) T v = (Tv)^\top (Y - X) (Tv) \geq 0$;
- (ii) $\lambda_{\min}(X)I \leq X \leq \lambda_{\max}(X)I$, and $\|X\| = \lambda_{\max}(X)$ if $X \geq 0$;
- (iii) if $0 < X \leq Y$ then $Y^{-1} \leq X^{-1}$;
- (iv) if $0 \leq X \leq Y$ then $\|X\| \leq \|Y\|$ and $\text{tr } X \leq \text{tr } Y$;
- (v) if $-aI \leq X \leq bI$ with $a, b \geq 0$ then $\|X\| \leq \max\{a, b\}$.

(F2) Discrete Lyapunov equation. Let F be Schur and D symmetric. Then $X = F^\top X F + D$ has the unique solution

$$X = \sum_{t=0}^{\infty} (F^\top)^t D F^t,$$

and $D \geq 0$ implies $X \geq 0$.

Proof. Substituting the equation into itself T times gives

$$\begin{aligned} X &= F^\top X F + D \\ &= F^\top (F^\top X F + D) F + D \\ &= (F^\top)^2 X F^2 + F^\top D F + D \\ &\vdots \\ &= (F^\top)^T X F^T + \sum_{t=0}^{T-1} (F^\top)^t D F^t. \end{aligned}$$

Since F is Schur, $F^T \rightarrow 0$, so the first term vanishes as $T \rightarrow \infty$ and the series converges. If X' is a second solution, then $Y = X - X'$ satisfies $Y = F^\top Y F$, so $Y = (F^\top)^T Y F^T \rightarrow 0$ and $Y = 0$. Each term of the series is ≥ 0 when $D \geq 0$ by (F1)(i). $\square$

(F3) Comparison principle. Let F be Schur, $X = F^\top X F + D$, and let Y be symmetric with $F^\top Y F + D \leq Y$. Then $X \leq Y$.

Proof. Subtracting the two relations gives

$$Y - X = F^\top (Y - X) F + \underbrace{(Y - F^\top Y F - D)}_{\geq 0}.$$

By (F2), $Y - X = \sum_{t=0}^{\infty} (F^\top)^t (Y - F^\top Y F - D) F^t \geq 0$. $\square$

(F4) Lyapunov stability test. If $P > 0$ and $F^\top P F - P < 0$, then F is Schur.

Proof. Let $Fv = \lambda v$ with $v \in \mathbb{C}^n$, $v \neq 0$. Then

$$v^* (F^\top P F - P) v = (|\lambda|^2 - 1) v^* P v.$$

The left-hand side is negative and $v^* P v > 0$, so $|\lambda| < 1$. $\square$

(F5) Kronecker identities.

- (i) $\text{vec}(XYZ) = (Z^\top \otimes X) \text{vec}(Y)$;
- (ii) $(X \otimes Y)(U \otimes V) = (XU) \otimes (YV)$;

- (iii) the eigenvalues of $X \otimes Y$ are the products $\lambda_i(X)\lambda_j(Y)$, so $\rho(X \otimes Y) = \rho(X)\rho(Y)$;
- (iv) $z^\top Wz = \text{vec}(W)^\top (z \otimes z)$, which is (i) with $X = z^\top$, $Y = W$, $Z = z$.

(F6) XY and YX . For $X \in \mathbb{R}^{d \times n}$ and $Y \in \mathbb{R}^{n \times d}$, XY and YX have the same nonzero eigenvalues.

Proof. Let $XYv = \lambda v$ with $\lambda \neq 0$, $v \neq 0$. Then $Yv \neq 0$, since otherwise $\lambda v = XYv = 0$. Moreover $YX(Yv) = Y(XYv) = \lambda Yv$, so λ is an eigenvalue of YX . The converse is symmetric. $\square$

(F7) Minimizing a quadratic in u . Let $H = \begin{bmatrix} H^{xx} & H^{xu} \\ H^{ux} & H^{uu} \end{bmatrix}$ be symmetric with $H^{uu} > 0$, and let $K^+ := (H^{uu})^{-1}H^{ux}$. Then for all x, u

$$\begin{bmatrix} x \\ u \end{bmatrix}^\top H \begin{bmatrix} x \\ u \end{bmatrix} = (u + K^+ x)^\top H^{uu} (u + K^+ x) + x^\top (H^{xx} - H^{xu} K^+) x,$$

so $\min_u [x; u]^\top H [x; u] = x^\top (H^{xx} - H^{xu} K^+) x$, attained only at $u = -K^+ x$.

Proof. Expand both sides and use $H^{uu} K^+ = H^{ux}$:

$$\begin{aligned} & (u + K^+ x)^\top H^{uu} (u + K^+ x) \\ &= u^\top H^{uu} u + 2u^\top H^{uu} K^+ x + x^\top K^{+\top} H^{uu} K^+ x \\ &= u^\top H^{uu} u + 2u^\top H^{ux} x + x^\top H^{xu} K^+ x. \end{aligned}$$

Adding $x^\top H^{xx} x - x^\top H^{xu} K^+ x$ gives $x^\top H^{xx} x + 2u^\top H^{ux} x + u^\top H^{uu} u = [x; u]^\top H [x; u]$. $\square$

(F8) Matrix Young inequality. For $P \geq 0$ and $\eta > 0$,

$$(X + E)^\top P (X + E) \leq (1 + \eta) X^\top P X + (1 + \eta^{-1}) E^\top P E.$$

Proof. By (F1)(i),

$$\begin{aligned} 0 &\leq (\sqrt{\eta} X - E / \sqrt{\eta})^\top P (\sqrt{\eta} X - E / \sqrt{\eta}) \\ &= \eta X^\top P X - X^\top P E - E^\top P X + \eta^{-1} E^\top P E, \end{aligned}$$

so $X^\top P E + E^\top P X \leq \eta X^\top P X + \eta^{-1} E^\top P E$. Insert this bound on the cross terms into

$$(X + E)^\top P (X + E) = X^\top P X + X^\top P E + E^\top P X + E^\top P E. \quad \square$$

Appendix A.2. Proof of Lemma 5

Write $\Pi = \Pi_K$, $G = G_K$, $P = P_K(W)$, $H = H_K(W)$.

Step 1 (spectral radius). Apply (F6) with $X = \Pi \in \mathbb{R}^{d \times n}$ and $Y = \Theta \in \mathbb{R}^{n \times d}$:

$$XY = \Pi\Theta = G, \quad YX = \Theta\Pi = A - BK.$$

The two matrices share their nonzero eigenvalues, so $\rho(G) = \rho(A - BK)$. In particular G is Schur if and only if $K \in \mathcal{K}(\Theta)$.

Step 2 (stage cost as a quadratic form). Since $W = \text{blkdiag}(Q, R)$,

$$\Pi^\top W \Pi = \begin{bmatrix} I & -K^\top \end{bmatrix} \begin{bmatrix} Q & 0 \\ 0 & R \end{bmatrix} \begin{bmatrix} I \\ -K \end{bmatrix} = Q + K^\top R K.$$

Step 3 ($P = \Pi^\top H \Pi$). Using $A - BK = \Theta\Pi$ and Step 2, the Lyapunov equation (3) becomes

$$\begin{aligned} P &= (\Theta\Pi)^\top P (\Theta\Pi) + \Pi^\top W \Pi \\ &= \Pi^\top (\Theta^\top P \Theta) \Pi + \Pi^\top W \Pi \\ &= \Pi^\top (W + \Theta^\top P \Theta) \Pi \\ &= \Pi^\top H \Pi. \end{aligned}$$

Step 4 (Lyapunov equation for H). Using $G = \Pi\Theta$ and Step 3,

$$\begin{aligned} G^\top H G &= \Theta^\top (\Pi^\top H \Pi) \Theta \\ &= \Theta^\top P \Theta \\ &= H - W, \end{aligned}$$

where the last equality is the definition (6). Hence $H = W + G^\top H G$.

Step 5 (uniqueness). G is Schur by Step 1, so (F2) gives uniqueness. $\square$

Appendix A.3. Proof of Proposition 6

Let $G = G_K$.

Step 1 (trajectory). Start at $z_0 = z = [x; u]$, i.e., take the action u first and follow K afterwards. Then $x_1 = \Theta z_0$ and $u_1 = -Kx_1$, so

$$z_1 = \begin{bmatrix} x_1 \\ -Kx_1 \end{bmatrix} = \Pi_K x_1 = \Pi_K \Theta z_0 = G z_0.$$

The same argument at every later step gives $z_t = G^t z$ by induction.

Step 2 (features along the trajectory). By (F5)(ii),

$$\begin{aligned} \phi(z_t) &= (G^t z) \otimes (G^t z) \\ &= (G^t \otimes G^t)(z \otimes z) \\ &= (G \otimes G)^t (z \otimes z). \end{aligned}$$

Step 3 (summing the series, part (a)). By (F5)(iii) and Lemma 5, $\rho(G \otimes G) = \rho(G)^2 < 1$. The Neumann series therefore converges:

$$\begin{aligned} \psi^K(z) &= \sum_{t \geq 0} \phi(z_t) = \left(\sum_{t \geq 0} (G \otimes G)^t \right) (z \otimes z) \\ &= (I - G \otimes G)^{-1} (z \otimes z). \end{aligned}$$

Each entry of $\psi^K(z)$ is a linear combination of the entries of $z \otimes z$, i.e., a quadratic form in z .

Step 4 (Q-function, part (b) first claim). The Q-function is the cost of the whole trajectory of Step 1. By (F5)(iv),

$$\begin{aligned} Q_K^W(z) &= \sum_{t \geq 0} z_t^\top W z_t \\ &= \sum_{t \geq 0} \text{vec}(W)^\top \phi(z_t) \\ &= \psi^K(z)^\top \text{vec}(W). \end{aligned}$$

Step 5 (successor operator, part (b) second claim). Vectorize $H_K = W + G^\top H_K G$ (Lemma 5) with (F5)(i), taking $X = G^\top$, $Y = H_K$, $Z = G$:

$$\text{vec}(H_K) = \text{vec}(W) + (G^\top \otimes G^\top) \text{vec}(H_K).$$

Hence $(I - G^\top \otimes G^\top) \text{vec}(H_K) = \text{vec}(W)$. The matrix on the left is invertible because $\rho(G^\top \otimes G^\top) = \rho(G)^2 < 1$, which gives $\text{vec}(H_K) = \mathcal{L}_K \text{vec}(W)$.

Step 6 (part (c)). $G = \Pi_K \Theta$ contains the system only through Θ , and so do $\mathcal{L}_K$ and ψ^K . $\square$

Appendix A.4. Proof of Theorem 8

Step 1 ($\hat{\Theta} = \Theta$). With $w_k \equiv 0$, every sample satisfies $x_{k+1} = \Theta z_k$. Stacking the samples gives $X^+ = \Theta Z$. PE means $ZZ^\top$ is invertible, so

$$\begin{aligned} \hat{\Theta} &= X^+ Z^\top (ZZ^\top)^{-1} \\ &= \Theta Z Z^\top (ZZ^\top)^{-1} \\ &= \Theta. \end{aligned}$$

Consequently $\hat{G}_K = G_K$, $\hat{H}_K = H_K$ and $\hat{P}_K = P_K$ for every K and every W . This step does not involve W .

Step 2 (the iteration is Hewer's algorithm). Suppose $K_j \in \mathcal{K}(\Theta)$; Step 4 below shows this holds for every j . By Lemma 5, the first equation in (14) returns $H_{K_j}(W)$. With $\Theta = [A \ B]$ and $P_j := P_{K_j}(W)$,

$$\Theta^\top P_j \Theta = \begin{bmatrix} A^\top P_j A & A^\top P_j B \\ B^\top P_j A & B^\top P_j B \end{bmatrix},$$

so the blocks of $H_{K_j} = W + \Theta^\top P_j \Theta$ are

$$H^{uu} = R + B^\top P_j B, \quad H^{ux} = B^\top P_j A.$$

The second equation in (14) therefore reads

$$K_{j+1} = (R + B^\top P_j B)^{-1} B^\top P_j A,$$

which is exactly Hewer's update Hewer (1971). Below, $S_j := R + B^\top P_j B > 0$, so that

$$B^\top P_j A = S_j K_{j+1}. \quad (\text{A.1})$$

Step 3 (a key identity). For an arbitrary gain L define

$$f(L) := (A - BL)^\top P_j (A - BL) + Q + L^\top R L.$$

Expanding the product term by term:

$$\begin{aligned} f(L) &= A^\top P_j A - A^\top P_j B L - L^\top B^\top P_j A \\ &\quad + L^\top B^\top P_j B L + Q + L^\top R L \\ &= A^\top P_j A + Q - A^\top P_j B L - L^\top B^\top P_j A \\ &\quad + L^\top S_j L. \end{aligned}$$

Substituting (A.1), i.e. $B^\top P_j A = S_j K_{j+1}$ and $A^\top P_j B = K_{j+1}^\top S_j$:

$$f(L) = A^\top P_j A + Q - K_{j+1}^\top S_j L - L^\top S_j K_{j+1} + L^\top S_j L.$$

Completing the square, using

$$\begin{aligned} (L - K_{j+1})^\top S_j (L - K_{j+1}) \\ = L^\top S_j L - L^\top S_j K_{j+1} - K_{j+1}^\top S_j L + K_{j+1}^\top S_j K_{j+1}, \end{aligned}$$

we obtain for every L :

$$\begin{aligned} f(L) &= A^\top P_j A + Q - K_{j+1}^\top S_j K_{j+1} \\ &\quad + (L - K_{j+1})^\top S_j (L - K_{j+1}). \end{aligned} \quad (\text{A.2})$$

Now evaluate (A.2) at two gains.

- $L = K_j$: $f(K_j) = P_j$ is the Lyapunov equation of P_j . Hence

$$P_j = A^\top P_j A + Q - K_{j+1}^\top S_j K_{j+1} + \Delta_j^\top S_j \Delta_j,$$

with $\Delta_j := K_j - K_{j+1}$.

- $L = K_{j+1}$: the last term vanishes, so

$$f(K_{j+1}) = A^\top P_j A + Q - K_{j+1}^\top S_j K_{j+1}.$$

Subtracting the second line from the first, and writing $A_{j+1} := A - B K_{j+1}$:

$$\begin{aligned} P_j &= A_{j+1}^\top P_j A_{j+1} + Q + K_{j+1}^\top R K_{j+1} \\ &\quad + \Delta_j^\top S_j \Delta_j. \end{aligned} \quad (\text{A.3})$$

Step 4 (part (a): every iterate is stabilizing). We argue by induction on j ; $K_0 \in \mathcal{K}(\Theta)$ by assumption. Assume $K_j \in \mathcal{K}(\Theta)$. Then $P_j \geq Q > 0$ by (F2). Write (A.3) as

$$A_{j+1}^\top P_j A_{j+1} - P_j = -(Q + K_{j+1}^\top R K_{j+1} + \Delta_j^\top S_j \Delta_j).$$

The right-hand side is $\leq -Q < 0$. By (F4), A_{j+1} is Schur, so $K_{j+1} \in \mathcal{K}(\Theta)$. In words: *the value matrix of the old gain is a Lyapunov function for the new gain.*

Step 5 (part (b): monotonicity). Since K_{j+1} is stabilizing, P_{j+1} solves

$$P_{j+1} = A_{j+1}^\top P_{j+1} A_{j+1} + Q + K_{j+1}^\top R K_{j+1}.$$

Subtracting this from (A.3):

$$P_j - P_{j+1} = A_{j+1}^\top (P_j - P_{j+1}) A_{j+1} + \Delta_j^\top S_j \Delta_j.$$

Since A_{j+1} is Schur and $\Delta_j^\top S_j \Delta_j \geq 0$, (F2) gives $P_j - P_{j+1} \geq 0$. The lower bound $P^* \leq P_{j+1}$ holds because, for every x_0 , $x_0^\top P^* x_0$ is the smallest achievable cost and $x_0^\top P_{j+1} x_0$ is the cost of one particular policy.

Step 6 (part (c): convergence). For every fixed v , the scalar sequence $v^\top P_j v$ is nonincreasing (Step 5) and bounded below by $v^\top P^* v$, so it converges. Taking $v = e_i$ and $v = e_i + e_l$ shows that every entry of P_j converges; let $\bar{P} := \lim_j P_j$. By continuity,

$$K_j \rightarrow \bar{K} := (R + B^\top \bar{P} B)^{-1} B^\top \bar{P} A.$$

Passing to the limit in (A.3), where $\Delta_j \rightarrow 0$, gives $\bar{P} = (A - B \bar{K})^\top \bar{P} (A - B \bar{K}) + Q + \bar{K}^\top R \bar{K}$. Since $\bar{K}$ is the improvement of

$\bar{P}$, identity (A.2) written with $\bar{P}$ in place of P_j and evaluated at $L = \bar{K}$ turns the right-hand side into

$$\begin{aligned}\bar{P} &= A^\top \bar{P} A + Q - \bar{K}^\top (R + B^\top \bar{P} B) \bar{K} \\ &= A^\top \bar{P} A + Q - A^\top \bar{P} B (R + B^\top \bar{P} B)^{-1} B^\top \bar{P} A,\end{aligned}$$

which is the DARE (5). Because $Q > 0$, $(A, Q^{1/2})$ is observable, and the DARE has exactly one positive semidefinite solution, namely P^* Lancaster and Rodman (1995). Since $\bar{P} \geq P^* \geq 0$, it follows that $\bar{P} = P^*$ and $\bar{K} = K^*$. Hwer's iteration is Newton's method applied to the Riccati operator, which gives the quadratic rate Hewer (1971); Lancaster and Rodman (1995).

Step 7 (equivalence with off-policy Q-learning). To evaluate K , off-policy Q-learning Bradtke et al. (1994); Kiumarsi et al. (2014) solves in the least-squares sense the N equations

$$z_k^\top H z_k - \tilde{z}_{k+1}^\top H \tilde{z}_{k+1} = z_k^\top W z_k, \quad \tilde{z}_{k+1} := \Pi_K x_{k+1}.$$

Without noise, $\tilde{z}_{k+1} = \Pi_K \Theta z_k = G_K z_k$, so the k -th equation reads

$$z_k^\top (H - G_K^\top H G_K - W) z_k = 0. \quad (\text{A.4})$$

- H_K is a solution. By Lemma 5, $H_K - G_K^\top H_K G_K - W = 0$, so every equation holds with zero residual.
- The solution is unique. Let H' be another symmetric solution, set $X := H' - H_K$, and let $Y := X - G_K^\top X G_K$ (symmetric). Subtracting (A.4) for H' and for H_K gives $z_k^\top Y z_k = 0$, i.e. $\langle Y, z_k z_k^\top \rangle = 0$ for all k , in the trace inner product. If the matrices $z_k z_k^\top$ span the symmetric matrices, then Y is orthogonal to every symmetric matrix, including itself, so $\langle Y, Y \rangle = 0$ and $Y = 0$. Thus $X = G_K^\top X G_K$, and iterating gives $X = (G_K^\top)^j X G_K^j \rightarrow 0$ because G_K is Schur. Hence $H' = H_K$.

Both methods therefore evaluate the same $H_{K_j}(W)$ and apply the same improvement formula. Starting from the same K_0 , their iterates coincide by induction.

Step 8 (backward map). Noise-free labels satisfy $r_k = z_k^\top W z_k = \phi(z_k)^\top \text{vec}(W)$. The pseudo-inverse in (12) returns the minimum-norm least-squares solution, which lies in the span of the vectors $\phi(z_k) = \text{vec}(z_k z_k^\top)$, so $\hat{w} = \text{vec}(\hat{W})$ with $\hat{W}$ symmetric. The residual is zero because w itself fits every label, hence $\langle \hat{W} - W, z_k z_k^\top \rangle = 0$ for all k . The spanning argument of Step 7 then gives $\hat{W} = W$. $\square$

Appendix A.5. Proof of Theorem 9

Write $Q_i := Q_{K_i}^W$, $P_i := P_{K_i}(W)$, $H_i := H_{K_i}(W)$ and $\ell(x, u) := x^\top Q x + u^\top R u$.

Step 1 (the library Q-functions are exact). Since $\hat{\Theta} = \Theta$ and $K_i \in \mathcal{K}(\Theta)$, Lemma 5 and Proposition 6 show that $\hat{\mathcal{L}}_{K_i} \text{vec}(W)$ returns the true kernel H_i . By Step 2 of the proof of Theorem 8, $H_i^{\text{true}} = R + B^\top P_i B > 0$.

Step 2 (closed form of GPI). By (F7) applied to H_i ,

$$\min_u Q_i(x, u) = x^\top S_i x,$$

with unique minimizer $u = -K_i^+ x$. Minimizing over i as well,

$$V(x) = \min_i x^\top S_i x = \min_i \min_u Q_i(x, u).$$

For $i = i^*(x)$ and $u = \pi_W^{\text{GPI}}(x) = -K_i^+ x$, both minima are attained, so

$$V(x) = Q_i(x, u). \quad (\text{A.5})$$

Step 3 (two Bellman identities). For every i and (x, u) , with $x^+ = Ax + Bu = \Theta z$:

$$\begin{aligned}Q_i(x, u) &= z^\top (W + \Theta^\top P_i \Theta) z \\ &= \ell(x, u) + x^{+\top} P_i x^+, \quad (\text{A.6})\end{aligned}$$

$$\begin{aligned}x^{+\top} P_i x^+ &= x^{+\top} \Pi_{K_i}^\top H_i \Pi_{K_i} x^+ \\ &= Q_i(x^+, -K_i x^+), \quad (\text{A.7})\end{aligned}$$

where (A.7) uses $P_i = \Pi_{K_i}^\top H_i \Pi_{K_i}$ (Lemma 5).

Step 4 (one-step decrease). Fix x , and let $i = i^*(x)$, $u = \pi_W^{\text{GPI}}(x)$, $x^+ = Ax + Bu$. Then

$$\begin{aligned}V(x) &= Q_i(x, u) && \text{by (A.5)} \\ &= \ell(x, u) + x^{+\top} P_i x^+ && \text{by (A.6)} \\ &= \ell(x, u) + Q_i(x^+, -K_i x^+) && \text{by (A.7)} \\ &\geq \ell(x, u) + \min_{u'} Q_i(x^+, u') && \min \leq \text{any value} \\ &= \ell(x, u) + x^{+\top} S_i x^+ && \text{by Step 2} \\ &\geq \ell(x, u) + V(x^+) && V \text{ is a min over } i.\end{aligned}$$

Therefore

$$V(x^+) \leq V(x) - \ell(x, u) \leq V(x) - x^\top Q x. \quad (\text{A.8})$$

The two inequalities are the two ways in which GPI does better than committing to K_i forever: it may re-choose the action (4th line) and the library policy (6th line) at the next step.

Step 5 (quadratic bounds on V). Lower bound. By (A.6) and Step 2,

$$x^\top S_i x = \min_u \{ \ell(x, u) + (Ax + Bu)^\top P_i (Ax + Bu) \} \geq x^\top Q x,$$

because the two terms not involving Q are nonnegative. Hence $S_i \geq Q$ and $V(x) \geq \lambda_{\min}(Q) \|x\|^2$. Upper bound. $V(x) \leq x^\top S_1 x \leq \bar{s} \|x\|^2$ with $\bar{s} := \max_i \|S_i\|$.

Step 6 (part (a): exponential stability). From (A.8), Step 5, and $\|x\|^2 \geq V(x)/\bar{s}$:

$$\begin{aligned}V(x^+) &\leq V(x) - \lambda_{\min}(Q) \|x\|^2 \\ &\leq V(x) - \frac{\lambda_{\min}(Q)}{\bar{s}} V(x) \\ &= \kappa V(x), \quad \kappa := 1 - \frac{\lambda_{\min}(Q)}{\bar{s}}.\end{aligned}$$

Since $S_i \geq Q$, $\bar{s} \geq \lambda_{\min}(Q)$, so $\kappa \in [0, 1)$. Iterating gives $V(x_t) \leq \kappa^t V(x_0)$, and by Step 5

$$\|x_t\|^2 \leq \frac{V(x_t)}{\lambda_{\min}(Q)} \leq \frac{\bar{s}}{\lambda_{\min}(Q)} \kappa^t \|x_0\|^2,$$

which is global exponential stability. Note that V is a minimum of quadratics and hence not smooth; this is harmless, since the discrete-time Lyapunov argument uses only the decrease inequality.

Step 7 (part (b): cost bound). By the first inequality of (A.8), $\ell(x_t, u_t) \leq V(x_t) - V(x_{t+1})$. Summing over $t = 0, \dots, T-1$, the right-hand side telescopes:

$$\sum_{t=0}^{T-1} \ell(x_t, u_t) \leq V(x_0) - V(x_T) \leq V(x_0).$$

Letting $T \rightarrow \infty$ bounds the accumulated cost by $V(x_0)$. Finally, taking $u = -K_i x$ in Step 2 and using (A.7),

$$x^\top S_i x \leq Q_i(x, -K_i x) = x^\top P_i x,$$

so $V(x_0) \leq \min_i x_0^\top P_i x_0$. $\square$

Appendix A.6. Proof of Lemma 10

We use the self-normalized martingale bound of (Abbasi-Yadkori et al., 2011, Theorem 1), restated here in our notation.

(F9) Self-normalized bound. Let $\{\mathcal{F}_k\}$ be a filtration, let $z_k \in \mathbb{R}^d$ be $\mathcal{F}_k$ -measurable, and let the scalar η_k be $\mathcal{F}_{k+1}$ -measurable with $\mathbb{E}[e^{s\eta_k} | \mathcal{F}_k] \leq e^{s^2 \sigma^2/2}$ for all $s \in \mathbb{R}$. Let $\Lambda > 0$ be deterministic, $\Gamma := \sum_{k \leq N} z_k z_k^\top$ and $S := \sum_{k \leq N} z_k \eta_k$. Then for every $\delta' \in (0, 1)$, with probability at least $1 - \delta'$,

$$S^\top (\Gamma + \Lambda)^{-1} S \leq 2\sigma^2 \log \frac{\det(\Gamma + \Lambda)^{1/2} \det(\Lambda)^{-1/2}}{\delta'}.$$

The name refers to the fact that S is normalized by its own random ‘‘covariance’’ Γ . A plain Hoeffding bound does not apply, because z_k contains x_k , which depends on the past noise; the terms $z_k \eta_k$ are not independent, but they form a martingale.

Step 1 (error formula). With $W_N := [w_0, \dots, w_{N-1}]$, the model (1) gives $X^+ = \Theta Z + W_N$. Hence

$$\begin{aligned} \hat{\Theta} - \Theta &= (\Theta Z + W_N) Z^\top (Z Z^\top)^{-1} - \Theta \\ &= W_N Z^\top (Z Z^\top)^{-1}. \end{aligned}$$

Step 2 (one row at a time). Let $\Gamma := Z Z^\top = N \hat{\Sigma}_{zz}$. Row i of W_N is $(w_{0,i}, \dots, w_{N-1,i})$, so row i of $W_N Z^\top$ is $(\sum_k w_{k,i} z_k)^\top =: S_i^\top$. Therefore

$$\text{row } i \text{ of } (\hat{\Theta} - \Theta) = S_i^\top \Gamma^{-1}, \quad i = 1, \dots, n.$$

Step 3 (apply (F9)). By Assumption 2, z_k is $\mathcal{F}_k$ -measurable and $\eta_k := w_{k,i}$ is conditionally σ_w -sub-Gaussian. Take the deterministic $\Lambda := N\lambda_0 I$ and $\delta' := \delta/n$. With probability at least $1 - \delta/n$,

$$S_i^\top (\Gamma + \Lambda)^{-1} S_i \leq 2\sigma_w^2 \log \frac{\det(\Gamma + \Lambda)^{1/2} \det(\Lambda)^{-1/2}}{\delta/n}. \quad (\text{A.9})$$

Step 4 (remove Λ using PE). On the event $\lambda_0 I \leq \hat{\Sigma}_{zz}$:

$$\begin{aligned} \Gamma &= N \hat{\Sigma}_{zz} \geq N\lambda_0 I = \Lambda \\ \Rightarrow \Gamma + \Lambda &\leq 2\Gamma \\ \Rightarrow \Gamma^{-1} &\leq 2(\Gamma + \Lambda)^{-1} \quad \text{by (F1)(iii)} \\ \Rightarrow S_i^\top \Gamma^{-1} S_i &\leq 2S_i^\top (\Gamma + \Lambda)^{-1} S_i. \end{aligned}$$

Step 5 (bound the determinant). Let $\mu_1, \dots, \mu_d$ be the eigenvalues of $\hat{\Sigma}_{zz}$; on the event $\hat{\Sigma}_{zz} \leq \bar{z}^2 I$, $\mu_j \leq \bar{z}^2$. The matrices $\Gamma + \Lambda = N(\hat{\Sigma}_{zz} + \lambda_0 I)$ and $\Lambda = N\lambda_0 I$ commute, so

$$\begin{aligned} \frac{\det(\Gamma + \Lambda)}{\det \Lambda} &= \prod_{j=1}^d \frac{N\mu_j + N\lambda_0}{N\lambda_0} \\ &= \prod_{j=1}^d \left(1 + \frac{\mu_j}{\lambda_0}\right) \leq \left(1 + \frac{\bar{z}^2}{\lambda_0}\right)^d. \end{aligned}$$

Taking log and combining with (A.9) and Step 4:

$$S_i^\top \Gamma^{-1} S_i \leq 4\sigma_w^2 \left(\frac{d}{2} \log \left(1 + \frac{\bar{z}^2}{\lambda_0}\right) + \log \frac{n}{\delta} \right). \quad (\text{A.10})$$

Step 6 (from the weighted norm to the row norm). Write $\Gamma^{-1} S_i = \Gamma^{-1/2} (\Gamma^{-1/2} S_i)$. Then

$$\begin{aligned} \|S_i^\top \Gamma^{-1}\|^2 &= (\Gamma^{-1/2} S_i)^\top \Gamma^{-1} (\Gamma^{-1/2} S_i) \\ &\leq \|\Gamma^{-1}\| \|\Gamma^{-1/2} S_i\|^2 \\ &= \frac{S_i^\top \Gamma^{-1} S_i}{\lambda_{\min}(\Gamma)} \leq \frac{S_i^\top \Gamma^{-1} S_i}{N\lambda_0}. \end{aligned}$$

Together with (A.10), each row satisfies

$$\|\text{row } i\|^2 \leq \frac{4\sigma_w^2}{N\lambda_0} \left(\frac{d}{2} \log \left(1 + \frac{\bar{z}^2}{\lambda_0}\right) + \log \frac{n}{\delta} \right).$$

Step 7 (all rows together). Each of the n bounds (A.9) fails with probability at most δ/n , so by the union bound all of them hold simultaneously with probability at least $1 - \delta$. On that event and on the PE event, summing the n row bounds and using $\|\cdot\| \leq \|\cdot\|_F$:

$$\|\hat{\Theta} - \Theta\|^2 \leq \|\hat{\Theta} - \Theta\|_F^2 = \sum_{i=1}^n \|\text{row } i\|^2 \leq \bar{\varepsilon}_N(\delta)^2.$$

Step 8 (the form of the statement). Steps 4–7 show: if the PE event holds and all n bounds (A.9) hold, then $\|\hat{\Theta} - \Theta\| \leq \bar{\varepsilon}_N$. Hence the event $\mathcal{E}$ (PE holds but $\|\hat{\Theta} - \Theta\| > \bar{\varepsilon}_N$) is contained in the event that some bound (A.9) fails, which has probability at most δ . $\square$

Remark 16. (F9) requires Λ , and hence λ_0 , to be fixed before the data are seen; $\bar{z}$ enters only through the PE event and may be chosen freely. If λ_0 is to be chosen from the data, apply the lemma on the grid $\lambda_0 \in \{2^{-l}\}_{l \geq 1}$ with confidence $\delta_l = \delta/(l(l+1))$. A union bound then covers all grid points at total confidence δ , at the price of an additional $\log(l(l+1))$ inside $\bar{\varepsilon}_N$.

Appendix A.7. Proof of Lemma 11

The idea is to use the value matrix P_1 of the nominal model as a Lyapunov function for the perturbed closed loop.

Step 0 (notation). Let $\Pi := \Pi_K$, the one-step cost $M := \Pi^\top W \Pi = Q + K^\top R K$, the closed loops $A_1 := \Theta_1 \Pi$ and $A_2 := \Theta_2 \Pi$, and

$$E := A_2 - A_1 = (\Theta_2 - \Theta_1) \Pi.$$

P_1 solves $P_1 = A_1^\top P_1 A_1 + M$. If $\varepsilon = 0$ there is nothing to prove, so let $\varepsilon > 0$.

Step 1 (four basic inequalities, all expressed in units of M).

- (i) $M \geq c \Pi^\top \Pi \geq cI$. The first inequality is $W \geq cI$ with (F1)(i); the second is $\Pi^\top \Pi = I + K^\top K \geq I$.
- (ii) $p_1 \geq c$, hence $\varepsilon \leq c/(6p_1) \leq 1/6$. Indeed $P_1 = A_1^\top P_1 A_1 + M \geq M \geq cI$, and then (F1)(iv).
- (iii) $E^\top P_1 E \leq \frac{p_1 \varepsilon^2}{c} M$. For every v ,

$$\begin{aligned} v^\top E^\top P_1 E v &\leq p_1 \|(\Theta_2 - \Theta_1) \Pi v\|^2 \\ &\leq p_1 \varepsilon^2 \|\Pi v\|^2 = p_1 \varepsilon^2 v^\top \Pi^\top \Pi v \\ &\leq \frac{p_1 \varepsilon^2}{c} v^\top M v \quad \text{by (i).} \end{aligned}$$

- (iv) $P_1 \leq \frac{p_1}{c} M$, because $P_1 \leq p_1 I$ by (F1)(ii) and $I \leq M/c$ by (i).

Step 2 (Young's inequality with $\eta = \varepsilon$). Apply (F8) with $X = A_1$:

$$A_2^\top P_1 A_2 \leq (1 + \varepsilon) A_1^\top P_1 A_1 + (1 + \varepsilon^{-1}) E^\top P_1 E.$$

The choice $\eta = \varepsilon$ balances the two prices: $(1 + \eta)$ inflates P_1 by $O(\eta)$, and $(1 + \eta^{-1})$ multiplies $E^\top P_1 E = O(\varepsilon^2)$, giving $O(\varepsilon^2/\eta)$. With $\eta = \varepsilon$ both are $O(\varepsilon)$.

Step 3 (expressing the bound through P_1 and M). Substitute $A_1^\top P_1 A_1 = P_1 - M$ and (iii), noting $(1 + \varepsilon^{-1})\varepsilon^2 = \varepsilon + \varepsilon^2$:

$$\begin{aligned} A_2^\top P_1 A_2 &\leq (1 + \varepsilon)(P_1 - M) + (\varepsilon + \varepsilon^2) \frac{p_1}{c} M \\ &= P_1 + \varepsilon P_1 - (1 + \varepsilon)M + (\varepsilon + \varepsilon^2) \frac{p_1}{c} M. \end{aligned}$$

Bound the term εP_1 with (iv), $\varepsilon P_1 \leq \varepsilon \frac{p_1}{c} M$:

$$\begin{aligned} A_2^\top P_1 A_2 &\leq P_1 - (1 + \varepsilon)M + (2\varepsilon + \varepsilon^2) \frac{p_1}{c} M \\ &= P_1 - \gamma M, \end{aligned}$$

where

$$\gamma := 1 + \varepsilon - (2\varepsilon + \varepsilon^2) \frac{p_1}{c}. \quad (\text{A.11})$$

Step 4 (γ is bounded away from zero). By (ii), $\varepsilon \leq 1/6$, so $\varepsilon^2 \leq \varepsilon/6$ and

$$2\varepsilon + \varepsilon^2 \leq \frac{13}{6}\varepsilon.$$

With $a := \frac{13}{6} \varepsilon \frac{p_1}{c}$, the assumption $\varepsilon \leq c/(6p_1)$ gives $a \leq \frac{13}{36}$. Dropping the term $+\varepsilon \geq 0$ in (A.11),

$$\gamma \geq 1 - a \geq \frac{23}{36} > 0.$$

Step 5 (K stabilizes Θ_2). From Step 3 and (i),

$$A_2^\top P_1 A_2 - P_1 \leq -\gamma M \leq -\gamma c I < 0.$$

Since $P_1 \geq cI > 0$, (F4) shows that A_2 is Schur, i.e. K stabilizes Θ_2 .

Step 6 (cost on Θ_2). Let P_2 be the value matrix on Θ_2 , so $P_2 = A_2^\top P_2 A_2 + M$, and let $Y := P_1/\gamma$. Dividing Step 3 by $\gamma > 0$:

$$A_2^\top Y A_2 \leq Y - M, \quad \text{i.e.} \quad A_2^\top Y A_2 + M \leq Y.$$

Thus Y is a super-solution of the Lyapunov equation of P_2 , and (F3) with $F = A_2$ gives

$$P_2 \leq Y = \frac{1}{\gamma} P_1.$$

Interpretation: the true cost is the sum of the one-step costs M along the trajectory; Y “pays” at least M at every step, so it pays at least the true total.

Step 7 (explicit constant). Using $\gamma \geq 1 - a$ and $a \leq \frac{13}{36}$:

$$\begin{aligned} \frac{1}{\gamma} &\leq \frac{1}{1 - a} = 1 + \frac{a}{1 - a} \leq 1 + \frac{36}{23} a \\ &= 1 + \frac{36}{23} \cdot \frac{13}{6} \cdot \frac{\varepsilon p_1}{c} = 1 + \frac{78}{23} \cdot \frac{\varepsilon p_1}{c} \\ &\leq 1 + 6 \frac{\varepsilon p_1}{c}. \end{aligned}$$

With Step 6 this proves $P_2 \leq (1 + 6\varepsilon p_1/c) P_1$. (The sharper constant $78/23 \approx 3.4$ may be used in place of 6.) $\square$

Appendix A.8. Proof of Theorem 12

Step 0 (one probabilistic event for all tasks). By Lemma 10, with probability at least $1 - \delta$ the following implication holds:

$$\lambda_0 I \leq \hat{\Sigma}_{zz} \leq \bar{z}^2 I \implies \|\hat{\Theta} - \Theta\| \leq \bar{\varepsilon}_N.$$

From now on we work on this event and assume the PE condition, so $\|\hat{\Theta} - \Theta\| \leq \varepsilon := \bar{\varepsilon}_N$. Everything below is deterministic. Since the event does not depend on W , the conclusions hold for all $W \in \mathcal{W}$ simultaneously; no union bound over tasks is needed. Fix W and write $c = c(W)$, $\hat{p} = \|\hat{P}\|$, $p^* = \|P^*\|$. Mode PI starts from $K_0 \in \mathcal{K}(\hat{\Theta})$, so Hewer's argument (Steps 2–6 of the proof of Theorem 8, which use only the model) applied to $\hat{\Theta}$ shows that its limit $\hat{K}$ stabilizes $\hat{\Theta}$ and that its value on $\hat{\Theta}$ is the stabilizing DARE solution $\hat{P}$ of $\hat{\Theta}$. We write $\ell(x, u) := x^\top Q x + u^\top R u$.

Part (a). Assume $\varepsilon \leq c/(12\hat{p})$.

Step 1 (from the model to the plant). Apply Lemma 11 with $\Theta_1 = \hat{\Theta}$, $\Theta_2 = \Theta$, $K = \hat{K}$, $P_1 = \hat{P}$. Its condition $\varepsilon \leq c/(6\hat{p})$ holds. Hence $\hat{K} \in \mathcal{K}(\Theta)$ and

$$P_{\hat{K}} \leq (1 + \alpha) \hat{P}, \quad \alpha := \frac{6\varepsilon \hat{p}}{c} \leq \frac{6}{12} = \frac{1}{2}. \quad (\text{A.12})$$

Step 2 (bound p^* by $\hat{p}$). Optimality of P^* gives $P^* \leq P_{\hat{K}}$. With (A.12) and (F1)(iv),

$$p^* \leq \|P_{\hat{K}}\| \leq (1 + \alpha) \hat{p} \leq \frac{3}{2} \hat{p}.$$

Step 3 (the reverse direction is also allowed). Lemma 11 from Θ to $\hat{\Theta}$ requires $\varepsilon \leq c/(6p^*)$. By Step 2,

$$\frac{c}{6p^*} \geq \frac{c}{6 \cdot \frac{3}{2} \hat{p}} = \frac{c}{9\hat{p}} \geq \frac{c}{12\hat{p}} \geq \varepsilon.$$

Step 4 (from the plant to the model). Apply Lemma 11 with $\Theta_1 = \Theta$, $\Theta_2 = \hat{\Theta}$, $K = K^*$, $P_1 = P^*$. Then K^* stabilizes $\hat{\Theta}$, and its value on $\hat{\Theta}$, denoted $P_{\hat{\Theta}}(K^*)$, satisfies

$$P_{\hat{\Theta}}(K^*) \leq (1 + \beta) P^*,$$

$$\beta := \frac{6\varepsilon p^*}{c} \leq \frac{9\varepsilon \hat{p}}{c} \leq \frac{9}{12} = \frac{3}{4}.$$

Step 5 (optimality on the model). $\hat{P}$ is optimal for $\hat{\Theta}$, and K^* is one stabilizing gain for $\hat{\Theta}$, so

$$\hat{P} \leq P_{\hat{\Theta}}(K^*) \leq (1 + \beta)P^*.$$

Step 6 (sandwich, first line of (17)). Chaining Step 2, (A.12) and Step 5:

$$P^* \leq P_{\hat{K}} \leq (1 + \alpha)\hat{P} \leq (1 + \alpha)(1 + \beta)P^*.$$

The factor satisfies, using $\beta \leq \frac{3}{4}$ for the product term,

$$\begin{aligned} (1 + \alpha)(1 + \beta) - 1 &= \alpha + \beta + \alpha\beta \\ &\leq \alpha + \beta + \frac{3}{4}\alpha \\ &\leq \left(6 + 9 + \frac{3}{4} \cdot 6\right) \frac{\varepsilon \hat{p}}{c} \\ &= 19.5 \frac{\varepsilon \hat{p}}{c} \leq 20 \frac{\varepsilon \hat{p}}{c}. \end{aligned}$$

Step 7 (bound on $\|\hat{P} - P^\|$).* From Step 5, $\hat{P} - P^* \leq \beta P^* \leq \beta p^* I$. From Step 6, $P^* - \hat{P} \leq P_{\hat{K}} - \hat{P} \leq \alpha \hat{P} \leq \alpha \hat{p} I$. By (F1)(v),

$$\|\hat{P} - P^*\| \leq \max\{\alpha \hat{p}, \beta p^*\}.$$

The two candidates are $\alpha \hat{p} = 6\varepsilon \hat{p}^2/c$ and $\beta p^* \leq \frac{9\varepsilon \hat{p}}{c} \cdot \frac{3}{2} \hat{p} = 13.5 \varepsilon \hat{p}^2/c$. Hence $\|\hat{P} - P^*\| \leq 14\varepsilon \hat{p}^2/c$.

Step 8 (cost-difference identity). Let $S^* := R + B^\top P^* B$ and $Q^*(x, u) := \ell(x, u) + (Ax + Bu)^\top P^* (Ax + Bu)$, whose kernel is $H^* = W + \Theta^\top P^* \Theta$. By (F7), the DARE, and the formula for K^* ,

$$Q^*(x, u) - x^\top P^* x = (u + K^* x)^\top S^* (u + K^* x). \quad (\text{A.13})$$

Let x_t be the true closed-loop trajectory under $\hat{K}$ from $x_0 = x$, with $u_t = -\hat{K}x_t$. By Step 1 it is stable, so $x_t \rightarrow 0$. Then:

$$\begin{aligned} x^\top P_{\hat{K}} x &= \sum_{t \geq 0} \ell(x_t, u_t), \\ x^\top P^* x &= \sum_{t \geq 0} (x_t^\top P^* x_t - x_{t+1}^\top P^* x_{t+1}), \end{aligned}$$

where the second sum telescopes to $x_0^\top P^* x_0$ because $x_t \rightarrow 0$. Subtracting and using $x_{t+1} = Ax_t + Bu_t$ and (A.13):

$$\begin{aligned} x^\top (P_{\hat{K}} - P^*) x &= \sum_{t \geq 0} (\ell(x_t, u_t) + x_{t+1}^\top P^* x_{t+1} - x_t^\top P^* x_t) \\ &= \sum_{t \geq 0} (Q^*(x_t, u_t) - x_t^\top P^* x_t) \\ &= \sum_{t \geq 0} x_t^\top (K^* - \hat{K})^\top S^* (K^* - \hat{K}) x_t. \end{aligned}$$

Step 9 (bound on $\|\hat{K} - K^\|$).* Every term of the last sum is nonnegative; keeping only $t = 0$ and using $S^* \geq R \geq cI$:

$$x^\top (P_{\hat{K}} - P^*) x \geq c \|\hat{K} - K^*\| x\|^2.$$

From Step 6, $0 \leq P_{\hat{K}} - P^* \leq 20 \frac{\varepsilon \hat{p}}{c} P^*$, so by (F1)(iv) and Step 2,

$$\|P_{\hat{K}} - P^*\| \leq 20 \frac{\varepsilon \hat{p}}{c} p^* \leq 30 \frac{\varepsilon \hat{p}^2}{c}.$$

Combining the two displays, $c\|(\hat{K} - K^*)x\|^2 \leq 30 \frac{\varepsilon \hat{p}^2}{c} \|x\|^2$ for all x . Hence $\|\hat{K} - K^*\|^2 \leq 30\varepsilon \hat{p}^2/c^2$, i.e. $\|\hat{K} - K^*\| \leq \frac{\hat{p}}{c} \sqrt{30\varepsilon}$.

Part (b). Assume $\varepsilon \leq c_W/(18\bar{p}_W)$ and fix $W \in \mathcal{W}$. Then $c \geq c_W$ and $p^* \leq \bar{p}_W$.

Step 10 (the model problem is well posed). Since

$$\varepsilon \leq \frac{c_W}{18\bar{p}_W} \leq \frac{c}{18p^*} \leq \frac{c}{6p^*},$$

Lemma 11 from Θ to $\hat{\Theta}$ with $K = K^*$ applies. Hence K^* stabilizes $\hat{\Theta}$, so the pair $(\hat{A}, \hat{B})$ with $\hat{\Theta} = [\hat{A} \ \hat{B}]$ is stabilizable, and since $Q > 0$ the DARE of $\hat{\Theta}$ has a stabilizing solution $\hat{P}$, to which Mode PI converges (Hewer's argument applied to $\hat{\Theta}$). Moreover

$$\hat{P} \leq P_{\hat{\Theta}}(K^*) \leq \left(1 + \frac{6\varepsilon p^*}{c}\right) P^* \leq \left(1 + \frac{6}{18}\right) P^* = \frac{4}{3} P^*,$$

so $\hat{p} \leq \frac{4}{3} p^* \leq \frac{4}{3} \bar{p}_W$.

Step 11 (the certificate holds automatically). Using Step 10, $c \geq c_W$ and $16 \leq 18$:

$$\frac{c}{12\hat{p}} \geq \frac{c_W}{12 \cdot \frac{4}{3} \bar{p}_W} = \frac{c_W}{16\bar{p}_W} \geq \frac{c_W}{18\bar{p}_W} \geq \varepsilon.$$

Step 12 (uniform rate). By Step 11, part (a) applies to W , so $P_{\hat{K}} \leq (1 + 20\varepsilon \hat{p}/c) P^*$. Taking the trace, which is monotone by (F1)(iv), and using $J_W = \text{tr } P$:

$$\begin{aligned} \frac{J_W(\hat{K}) - J_W^*}{J_W^*} &\leq \frac{20\varepsilon \hat{p}}{c} \leq 20\varepsilon \cdot \frac{\frac{4}{3} \bar{p}_W}{c_W} \\ &= \frac{80}{3} \frac{\bar{p}_W}{c_W} \varepsilon \leq 27 \frac{\bar{p}_W}{c_W} \bar{\varepsilon}_N. \end{aligned}$$

The right-hand side does not depend on W , which gives (18). Substituting the definition (16) of $\bar{\varepsilon}_N$ gives the $O(\cdot)$ form. $\square$

Appendix A.9. Proof of Theorem 15

Let K_j be a gain with $A_j := A - BK_j$ Hurwitz, let $Q_j := Q + K_j^\top RK_j$, let P_j solve $A_j^\top P_j + P_j A_j + Q_j = 0$, and let $K_{j+1} := R^{-1} B^\top P_j$ (Kleinman's step).

Step 1 (the Kleinman pair satisfies (20)). Along the data trajectory, rewrite the dynamics as

$$\dot{x} = A_j x + B(u + K_j x).$$

Differentiating $x^\top P_j x$ and using the Lyapunov equation and $B^\top P_j = RK_{j+1}$:

$$\begin{aligned} \frac{d}{dt}(x^\top P_j x) &= x^\top (A_j^\top P_j + P_j A_j) x + 2(u + K_j x)^\top B^\top P_j x \\ &= -x^\top Q_j x + 2(u + K_j x)^\top RK_{j+1} x. \end{aligned}$$

Integrating over $[t_{i-1}, t_i]$:

$$\begin{aligned} x^\top P_j x \Big|_{t_{i-1}}^{t_i} &= - \int_{t_{i-1}}^{t_i} x^\top Q_j x d\tau \\ &\quad + 2 \int_{t_{i-1}}^{t_i} (u + K_j x)^\top R K_{j+1} x d\tau. \end{aligned} \quad (\text{A.14})$$

Each term is linear in the unknowns. By definition of $\bar{x}$, $x^\top P_j x = \bar{x}^\top \text{svec}(P_j)$. By (F5)(iv), $x^\top Q_j x = (x \otimes x)^\top \text{vec}(Q_j)$. By (F5)(i) with $X = (u + K_j x)^\top R$, $Y = K_{j+1}$, $Z = x$:

$$\begin{aligned} &(u + K_j x)^\top R K_{j+1} x \\ &= (x^\top \otimes (u + K_j x)^\top R) \text{vec}(K_{j+1}) \\ &= [(x^\top \otimes x^\top)(I_n \otimes K_j^\top R) + (x^\top \otimes u^\top)(I_n \otimes R)] \text{vec}(K_{j+1}), \end{aligned}$$

where the last line uses (F5)(ii) in the form $x^\top \otimes x^\top K_j^\top R = (x^\top \otimes x^\top)(I_n \otimes K_j^\top R)$, and similarly for u . Substituting into (A.14) and stacking the l intervals gives exactly (20), with zero residual.

Step 2 (the equation depends on data only through D). Let $q := \frac{n(n+1)}{2} + nm$ and

$$\begin{aligned} T(K_j, R) &:= \begin{bmatrix} I & 0 \\ 0 & -2(I_n \otimes K_j^\top R) \\ 0 & -2(I_n \otimes R) \end{bmatrix} \in \mathbb{R}^{p \times q}, \\ s(K_j, W) &:= \begin{bmatrix} 0 \\ -\text{vec}(Q_j) \\ 0 \end{bmatrix} \in \mathbb{R}^p, \end{aligned}$$

with row blocks of sizes $\frac{n(n+1)}{2}$, n^2 and nm matching $D = [\delta_{xx}, I_{xx}, I_{xu}]$. Multiplying out,

$$\begin{aligned} DT &= [\delta_{xx}, -2I_{xx}(I \otimes K_j^\top R) - 2I_{xu}(I \otimes R)], \\ Ds &= -I_{xx} \text{vec}(Q_j), \end{aligned}$$

so (20) is $DT\theta = Ds$, where T and s depend only on (K_j, W) .

Step 3 (normal equations use only the Gram matrix). Multiplying $DT\theta_j^* = Ds$ on the left by $T^\top D^\top$:

$$T^\top (D^\top D) T \theta_j^* = T^\top (D^\top D) s,$$

so the Kleinman pair θ_j^* solves the equation of Algorithm 3, which involves only $\mathcal{R}_{CT} = D^\top D$.

Step 4 (uniqueness). $T^\top D^\top DT$ is invertible if and only if DT has full column rank. We show that $DT\theta = 0$ implies $\theta = 0$, following the argument of Jiang and Jiang (2012). Let $\theta = [\text{svec}(X); \text{vec}(Y)]$ with X symmetric and $DT\theta = 0$. Along the data,

$$\frac{d}{dt}(x^\top X x) = x^\top (A^\top X + X A)x + 2u^\top B^\top X x.$$

Row i of $DT\theta = 0$ therefore reads

$$\begin{aligned} 0 &= \int [x^\top (A^\top X + X A)x + 2u^\top B^\top X x] d\tau \\ &\quad - 2 \int [x^\top K_j^\top R Y x + u^\top R Y x] d\tau \\ &= \int x^\top (A^\top X + X A - K_j^\top R Y - Y^\top R K_j) x d\tau \end{aligned}$$

$$+ 2 \int u^\top (B^\top X - R Y) x d\tau.$$

The first integral is a row of I_{xx} applied to a symmetric matrix, the second a row of I_{xu} applied to $\text{vec}(B^\top X - R Y)$. The rank condition states exactly that these columns are linearly independent, so

$$B^\top X = R Y, \quad A^\top X + X A - K_j^\top R Y - Y^\top R K_j = 0.$$

Substituting $R Y = B^\top X$ into the second equation gives

$$(A - B K_j)^\top X + X(A - B K_j) = 0.$$

Since $A - B K_j$ is Hurwitz, this Lyapunov equation has only the solution $X = 0$. Then $Y = R^{-1} B^\top X = 0$, so $\theta = 0$.

Step 5 (induction and convergence). By Steps 3–4, if K_j is stabilizing then Algorithm 3 returns exactly Kleinman's K_{j+1} . Kleinman's theorem Kleinman (1968) states that from a stabilizing K_0 every iterate is stabilizing, P_j decreases monotonically, and $K_j \rightarrow K^*(W)$ quadratically. By induction on j the two sequences coincide. The same K_0 serves every W , because stability of $A - B K_0$ does not involve W .

Step 6 (cost). Forming $T^\top \mathcal{R}_{CT} T$ with $T \in \mathbb{R}^{p \times q}$ and solving the $q \times q$ system costs $O(p^2 q + p q^2 + q^3) = O(p^3)$, independent of the number of intervals l . $\square$

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